Financial Flexibility under Non-Exclusive Lending

Yunzhi Hu and Felipe Varas*

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Abstract

This paper studies optimal financial flexibility in debt financing. A borrower raises funds for an initial investment and may need financing after observing a private liquidity shock. Under non-exclusive lending, the price of new debt does not internalize the dilution of existing creditors, leading to excessive leverage. We characterize conditions under which the optimal simple long-term debt contract limits dilution by imposing a debt limit. The optimal contract jointly determines the amount of initial debt and future borrowing discretion. A higher investment requirement increases initial debt and lowers the total debt limit. We show that richer clauses, such as performance-sensitive debt and contingent prepayment provisions, allow the implementation of the exclusive-lending outcome by compensating existing lenders when new financing is raised.

Keywords: financial flexibility, debt dilution, lending non-exclusivity, covenant, optimal delegation

*Hu: Kenan-Flagler Business School, UNC Chapel Hill; Yunzhi_Hu@kenan-flagler.unc.edu. Varas: Naveen Jindal School of Management, University of Texas at Dallas; felipe.varas@utdallas.edu. We are grateful to Jesse Davis, Alessandro Dovis, Paolo Fulghieri, Brett Green, Zhiguo He, Ron Kaniel, Narayana Kocherlakota, Adriano Rampini, Jacob Sagi, Fenghua Song (discussant), Andy Skrzypacz, Joao Thereze, Vivian Yue, Neng Wang, Pavel Zryumov and seminar participants from UNC, WashU Olin Finance Conference, 14th Advances in Macro-Finance Conference, Rochester Simon, and Finance Theory Group Meeting for useful comments.

1 Introduction

Financial flexibility is the ability of a firm to obtain financing when it needs it. Graham (2022) reports that financial flexibility is one of the most commonly cited consideration for debt policy, with 87% of large firms and 78% of small firms rate it as important or very important in their CFO survey. Firms value this flexibility because future financing needs are difficult to predict: new investment opportunities may arise, cash flows may deteriorate, or financial distress may force the firm back to capital markets.

However, preserving the ability to issue debt in the future can come at a cost.[1] In the context of debt financing, a firm that can freely borrow in the future may dilute the claims of existing creditors. A recent literature in dynamic capital structure has shown how this can induce firms to increase leverage even when doing so is suboptimal, thereby reducing the value from borrowing (Admati, Demarzo, Hellwig, and Pfleiderer, 2018; DeMarzo and He, 2021). Optimal debt policy must therefore balance future borrowing flexibility against the protection of existing creditors.

To study the trade-off between financial flexibility and creditor protection, we develop a three-date model in which a borrower initially issues debt to finance a fixed investment. The final outcome of the project depends on the borrower effort, so higher amounts of debt reduces effort incentives. In the interim period, the borrower privately observes a liquidity shock, modeled as a discount-factor shock, that creates an incentive to increase borrowing. Borrowing is non-exclusive: the borrower can turn to new lenders who only price the credit risk in the new claims, so the borrower does not internalize the dilution on existing claims, which generates incentives for over-borrowing.

To protect against dilution, the initial debt contract can restrict future borrowing. We show that the optimal financing problem is isomorphic to a delegation problem in which the borrower jointly chooses the initial level of debt and the set of future debt choices available at the interim date. The initial restrictions on future debt issuance anticipate that at the interim date, the borrower will choose additional borrowing

[1] A large literature studies the costs of flexibility that come from excess cash or discretion over internal resources, including the free-cash-flow problem in Jensen (1986) and related contracting problems such as Hart and Tirole (1988). Unlike this previous literature, our focus is to study flexibility in debt financing.

to maximize the continuation value net of the legacy debt claim, leading to a time-inconsistency problem. We show that if initial lenders can restrict future debt choices, but cannot use state-contingent transfers that compensate them for dilution, then under some technical conditions the optimal simple long-term debt contract is a limit on future debt issuance.

Initial debt is necessary to provide funding for initial investment, but it also determines the value of the legacy claim, and so the incentives to dilute existing creditors. Hence, the amount of future borrowing discretion the contract should allow must be jointly determined with the choice of initial debt. The solution contains both a within-period and an intertemporal commitment-flexibility tradeoff. For a given initial debt level, borrowers with severe liquidity shocks are constrained by the borrowing limit and under-borrow relative to the efficient allocation, while borrowers with mild shocks remain unconstrained and over-borrow. Across periods, we show that a higher investment requirement increases initial debt and lowers the optimal borrowing limit.

We compare the allocation with the optimal contract under exclusive lending, where the borrower must return to the initial lender for refinancing. In this case, the lender internalizes the effect of new borrowing, eliminating creditor dilution. Borrowers with sufficiently mild liquidity shocks receive zero transfers at refinancing. This happens because the limited liability constraint binds for these types, and incentive compatibility then requires the same total debt across these shocks. This leads to a stark contrast between the exclusive and non-exclusive lending allocations: under exclusive lending, the allocation has pooling for mild shocks (high discount factor), while under non-exclusive lending with simple debt contract the allocation has pooling for sufficiently severe liquidity shocks (low discount factor).

We then consider implementation under non-exclusivity with richer contractual feature, such as performance-sensitive debt and contingent prepayment provisions. We show how these richer contractual features can be designed to make the borrower internalize dilution, and hence implement the exclusive-lending allocation in the absence of exclusivity.

Although our main focus is the design of creditor protection, our approach and many of our conclusions extend to other contracting problems with non-exclusivity. We discuss related commitment problems in sovereign borrowing and durable goods in Section 6 to illustrate how the delegation approach can be applied to these settings.

Related Literature

Our paper is most closely related to the mechanism-design literature on commitment and flexibility using tools from the theory of delegation (Holmström, 1984). This literature studies how a principal delegates choices to an agent who posses useful information but whose preferences are misaligned with the principal. Amador, Werning, and Angeletos (2006) use this setting to study consumption-saving choices when the preference misalignment arises from time-inconsistency due to hyperbolic preferences. Since their seminal contribution, the delegation approach has been applied to study fiscal rules when governments privately observe their spending needs and political constraints lead to time-inconsistent preferences (Halac and Yared, 2014, 2018, 2022, 2025). In this paper, we build on the Lagrangian approach developed by Amador et al. (2006), and extended by Amador and Bagwell (2013) to a general class of delegation problems.

However, unlike this previous literature, which takes the disagreement between principal and agent as given, the disagreement in our problem comes from the initial choice of financing through long-term debt. Initial debt raises funds for investment and creates incentives to dilute existing lenders, while future borrowing discretion affects the price of the initial claim. The initial claim and the delegation set must therefore be chosen jointly. A higher investment requirement raises initial debt and lowers the total debt limit whenever some flexibility remains. The room for subsequent borrowing therefore shrinks both because more debt is issued initially and because the contract imposes a lower limit on future borrowing.

Our paper is also related to contracting under non-exclusivity, where a contracting party cannot prevent its counterparty from trading with others. In moral-hazard models, additional contracts weaken effort incentives by diluting existing lenders' claims (Kahn and Mookherjee, 1998; Bisin and Guaitoli, 2004; Bizer and DeMarzo, 1992). In adverse-selection models, non-exclusive trade complicates screening when privately informed sellers can issue securities to multiple buyers (Attar, Mariotti, and Salanié, 2011, 2014, 2020; Asriyan and Vanasco, 2024).

Creditor dilution is a particular example of contracting externalities. Borrowing from new lenders lowers initial lenders' payoffs, as in Segal (1999), who studies contracts in which one agent's trade affects other agents' payoffs. Unlike much of this literature, which takes non-exclusivity to mean that outside contracts cannot be restricted, we allow the initial contract to limit the borrower's aggregate outside

borrowing. So, our setting can be interpreted as a model of sequential contracting with externalities where contracts can limit the quantities traded with future agents.

With simple contracts, the principal can only limit new trading. However, once richer contracts are considered, the initial claim can respond to new issuance, following the contract-contingent logic of Peters and Szentes (2012) and Szentes (2015). We show that such state-contingent compensation to existing creditors allows to implement the exclusive-lending allocation without exclusivity.

Within corporate finance, the closest work to ours studies financing when borrowers cannot commit to future borrowing. Fama and Miller (1972) and Black and Scholes (1973) first highlighted that borrowers can increase leverage by issuing new debt without internalizing the resulting dilution of outstanding debt. Subsequent work studies how the debt-dilution problem shapes borrower incentives, leverage dynamics, and the value of commitment (Bizer and DeMarzo, 1992; Admati et al., 2018; DeMarzo, 2019; DeMarzo and He, 2021). In these models, lenders know the value of additional borrowing, so they could limit inefficient leverage. Our lenders cannot separate it from borrowing the firm needs, so a limit that prevents dilution might inefficiently restrict borrowers with severe shocks.

In practice, covenants help protect existing creditors against dilution from later financing (Smith and Warner, 1979). Attar, Casamatta, Chassagnon, and Décamps (2019) study how covenants restrict side trading and reduce lender competition under non-exclusive lending. Donaldson, Gromb, and Piacentino (2025) study anti-dilution covenants that balance creditor protection against valuable later financing. In their model, creditors observe project quality and can waive covenants when later financing creates value. In our model, the borrower privately observes its liquidity shock, so lenders cannot waive the limit even when it constrains the borrower inefficiently.

2 Model

We consider a three-date financing problem with both moral hazard and private information. A borrower raises funds for an initial investment at $t = 0$, learns a privately observed discount-factor shock at $t = 1$, and privately chooses effort at $t = 2$.

2.1 Borrower, Investment, and Discount-Factor Shock

There are three dates, $t = 0, 1, 2$, and a risk-neutral borrower. The borrower has a project that requires a fixed investment of $I > 0$ at $t = 0$ but has no resources to finance it, hence the need to borrow. The project generates no cash flow at $t = 1$ and a final cash flow Y at $t = 2$ if successful (zero otherwise).[2] At the beginning of $t = 2$, the borrower privately chooses an effort level $q \in [0, 1]$, where q corresponds to the success probability. The effort incurs a private cost $c(q)$. We assume that $c(0) = c'(0) = 0$ and $c'(1) > Y$, that $c(q)$ is twice continuously differentiable, and that $c''(q) > 0$ for every $q \in (0, 1]$. These conditions imply that effort is uniquely determined and interior.

The borrower does not discount cash flows between $t = 0$ and $t = 1$. At $t = 1$, the borrower receives a discount factor shock and discounts cash flows at $t = 2$ at θ. Only the borrower observes the realization of θ. We assume that θ is drawn from a set $\Theta = [\underline{\theta}, \bar{\theta}] \subseteq (0, 1]$, with a continuously differentiable density function $g(\theta) > 0$ and cumulative density function $G(\theta)$. Throughout the paper, we make the following standard assumption, which leads to the result that $g(\theta)/G(\theta)$ is decreasing.

Assumption 1. *The cumulative density function $G(\theta)$ is log-concave.*

2.2 Lenders and Contract

There exists a competitive pool of dispersed lenders. Throughout the paper, lenders are risk-neutral and do not discount future cash flows. Given that lenders are competitive, the borrower captures all surplus.

The contract specifies how the cash flow at $t = 2$ is split among the borrower, the initial lenders at $t = 0$, and any new lenders at $t = 1$. At $t = 0$, initial lenders provide an upfront transfer τ_0 to finance the investment I and receive a claim x_0 on the final cash flow. At $t = 1$, the borrower returns to the market and reports $\hat{\theta}$. New lenders provide an interim transfer $\tau_1(\hat{\theta})$ in exchange for the additional claim $x_1(\hat{\theta}) - x_0$, so that total promised repayment becomes $x_1(\hat{\theta})$. Note that at $t = 1$, x_0 is a legacy

[2] Under this binary cash flow framework, there is no fundamental distinction between different securities, such as equity and debt. In addition, it implies that seniority is irrelevant within the class of debt contracts. We choose to model final cash flows in this simple way to eliminate complicated security design considerations.

claim that can be diluted by additional borrowing. We interpret these claims as debt contracts and refer to x_1 as the face value of total debt. For a given realization of the discount-factor shock θ, report $\hat{\theta}$, and effort q, the borrower's payoff is

$$\tau_0 - I + \tau_1(\hat{\theta}) + \theta \left[q \left(Y - x_1(\hat{\theta}) \right) - c(q) \right].$$

Since the borrower has no initial wealth and is protected by limited liability, the contract must satisfy the following financing and feasibility constraints:

$$\tau_0 \geq I, \qquad \tau_1(\hat{\theta}) \geq 0, \qquad 0 \leq x_0 \leq x_1(\hat{\theta}) \leq Y \quad \text{for all } \hat{\theta} \in \Theta. \tag{1}$$

For the rest of the paper, we refer to the constraint $\tau_0 \geq I$ as the *financing constraint*.

Under non-exclusive lending, the borrower can use different lenders at $t = 0$ and $t = 1$. It may find it difficult to reassemble dispersed initial lenders at refinancing. Even relationship banks may face regulatory capital constraints that limit additional exposure to a borrower or sector. Moreover, as we show later, the borrower also prefers to borrow from new lenders when it can choose.

Figure 1 summarizes the sequence of events on three dates.

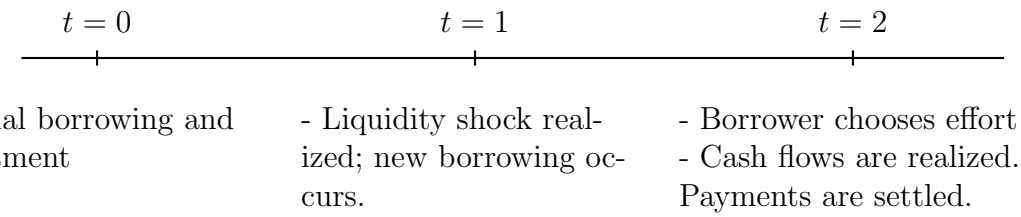


Figure 1: Timeline of events

2.3 Effort Choice and Valuations

We now turn to the borrower's problem at $t = 2$. The key tradeoff is moral hazard: higher debt weakens the borrower's incentive to exert effort.[3] Given a face value of

[3]The solution without a moral hazard problem is trivial: the borrower always pledges all the cash flows at $t = 0$. An alternative to introducing moral hazard is to assume bankruptcy costs.

debt x_1, the borrower chooses effort q to maximize its payoff

$$V_2(x_1) = \max_{q \in [0,1]} \; q(Y - x_1) - c(q), \tag{2}$$

where $V_2(x_1)$ is the value of the borrower's residual claim net of effort cost. Because there is no outsider equity, $\theta V_2(x_1)$ is also the firm's equity value at $t = 1$. Let $q(x_1)$ denote the optimal effort choice. The following lemma describes properties of $V_2(x_1)$ and $q(x_1)$ used throughout the paper.

Lemma 1. *On $[0, Y]$, optimal effort $q(x_1) = (c')^{-1}(Y - x_1)$ is strictly decreasing, while $V_2(x_1)$ is decreasing and convex with derivative $V_2'(x_1) = -q(x_1)$.*

For $x_1 \in [0, Y)$, $q'(x_1) = -1/c''(q(x_1)) < 0$. If c is three times continuously differentiable, then, for $x_1 \in (0, Y)$,

$$q''(x_1) = -\frac{c'''(q(x_1))}{c''(q(x_1))}\big(q'(x_1)\big)^2.$$

The condition $c'(0) = 0$ implies that $q(x_1) > 0$ and $V_2(x_1) > 0$ whenever $x_1 < Y$, while $q(Y) = V_2(Y) = 0$. We impose two assumptions. Assumption 2 states that the maximum expected repayment $x_1 q(x_1)$ is sufficient to cover the initial investment I. Assumption 3 and Lemma 1 imply that $q(x_1)$ is concave. This assumption will guarantee that the objective function of several of the programs we later consider is concave.

Assumption 2. *The cost function $c(q)$ satisfies*

$$I^{\max} \coloneqq \max_{x_1 \in [0,Y]} \; x_1 q(x_1) > I. \tag{3}$$

Assumption 3. *$c(q)$ is three times continuously differentiable and satisfies $c'''(q) \geq 0$.*

For the rest of this paper, we define

$$W_1(x_1, \theta) \coloneqq \underbrace{x_1 q(x_1)}_{\text{debt}} + \underbrace{\theta V_2(x_1)}_{\text{equity}}, \tag{4}$$

as the (ex-post) $t = 1$ firm value for a given θ, and we let $x_1^*(\theta)$ be the level of debt that maximizes it,

$$x_1^*(\theta) = \arg\max_{x_1 \in [0,Y]} \; W_1(x_1, \theta). \tag{5}$$

We call $x_1^*(\theta)$ the *ex post-efficient* debt level. It maximizes total surplus after observing θ when moral hazard is the only friction, balancing liquidity against effort incentives. A more patient borrower places less value on immediate liquidity, so efficient debt decreases with θ. Before observing the shock at $t = 0$, the borrower instead chooses $x_1^*(\mathbb{E}[\theta])$, which we call the *ex-ante optimal* debt. This follows because $W_1(x_1, \theta)$ is linear in θ.

Throughout the paper, we use the following linear example to illustrate the main results.

Linear Example. *In the linear example, the effort cost is given by the quadratic function,* $c(q) = \frac{1}{2}cq^2$*, with* $c > Y$*.*[4] *Effort is then linear in debt, and the ex post-efficient debt level is*

$$q(x_1) = \frac{Y - x_1}{c}, \qquad x_1^*(\theta) = \frac{1-\theta}{2-\theta}Y.$$

3 Non-Exclusive Lending with Simple Debt Contracts

Initial lenders anticipate dilution from new borrowing at $t = 1$ and seek to limit issuance. We consider contracts $\{\tau_0, x_0, x_1(\hat{\theta})\}$ offered to initial lenders, where $x_1(\hat{\theta})$ limits total debt as a function of the borrower's reported type. At $t = 1$, the contract rules out transfers to initial lenders and adjustments to x_0 in response to the report. We refer to this type of contracts financing contracts as simple debt contracts. Later in Section 5 we consider the impact of using more complex contracts where the terms of the initial contract can adjusted in reaction to new issuance.

Since the borrower faces a liquidity shock, we can assume without loss of generality that it issues new debt up to the debt limit $x_1(\hat{\theta})$, so $\Delta x_1(\hat{\theta}) \coloneqq x_1(\hat{\theta}) - x_0$ is the additional debt issued at $t = 1$.

[4] In the numerical illustrations, we use $c = Y = 1$. The formulas apply to this case, given $0 < x_1 < Y$.

3.1 Reduction to Delegation Problem

We first show that the problem can be equivalently formulated as a delegation problem. Because the borrower offers the contract, new lenders' participation constraint must bind, implying that $\tau_1 = (x_1 - x_0)q(x_1)$, where $x_1 \geq x_0$ is required to satisfy the non-negativity constraint $\tau_1 \geq 0$. At $t = 1$, the borrower's continuation payoff for any given (θ, x_1) is:

$$V_1(x_1, x_0, \theta) = \underbrace{(x_1 - x_0)q(x_1)}_{\text{new debt issuance}} + \theta V_2(x_1) = W_1(x_1, \theta) - x_0 q(x_1). \tag{6}$$

As the borrower has all the bargaining power at the initial date, the lenders' participation constraint binds, so $\tau_0 = x_0 \mathbb{E}[q(x_1(\theta))]$. The financing constraint therefore becomes

$$x_0 \mathbb{E}\left[q(x_1(\theta))\right] \geq I. \tag{7}$$

Substituting the lenders' break-even conditions into the borrower's payoff gives the objective $\mathbb{E}[W_1(x_1(\theta), \theta)] - I$. Since the investment level I is fixed, the borrower maximizes expected firm value. At $t = 1$, the borrower has incentives to report its type truthfully only if the following incentive compatibility constraint is satisfied

$$\theta \in \arg\max_{\hat{\theta} \in \Theta} \; V_1(x_1(\hat{\theta}), x_0, \theta). \tag{8}$$

By the envelope theorem, this incentive compatibility constraint can be rewritten as

$$\begin{aligned} &V_1(x_1(\theta), x_0, \theta) = V_1(x_1(\bar{\theta}), x_0, \bar{\theta}) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))d\tilde{\theta} \quad \forall \theta \in \Theta \\ &x_1(\theta) \text{ is non-increasing.} \end{aligned} \tag{9}$$

The optimal mechanism at $t = 0$ solves the following problem:

$$\begin{aligned} &\max_{x_0, x_1(\theta) \geq x_0} \mathbb{E}\Big[W_1(x_1, \theta)\Big] \\ &\text{subject to (7) and (9).} \end{aligned} \tag{10}$$

This is a delegation problem in which the borrower at $t = 0$ is the principal and the borrower at $t = 1$ is the agent. The principal delegates the choice of total debt to the agent, who privately observes the refinancing need θ. The principal maximizes firm

value net of investment I, whereas the agent maximizes firm value net of initial debt, $V_1(x_1, x_0, \theta)$. The conflict between the ex-ante objective and the ex-post objective relates to the insight in Aguiar, Amador, Hopenhayn, and Werning (2019), who show that the equilibrium debt issuance decisions with long-term debt can be characterized as the solution to a planner's problem that ignores the payoff to existing creditors.

Two features distinguish this problem from the previous literature on delegation. First, the financing constraint (7) couples the initial choice x_0 to the continuation allocation. Second, the disagreement arises from the initial debt the principal chooses, rather than from a preference parameter. The difference between the principal's and the agent's payoffs equals the value of that debt: $W_1(x_1, \theta) - V_1(x_1, x_0, \theta) = x_0 q(x_1)$. Thus, x_0 simultaneously relaxes the financing constraint and increases the wedge between the two objective functions. This is the source of the intertemporal commitment-flexibility tradeoff.

The optimimization problem (10) is not jointly convex in x_0 and $x_1(\theta)$, so neither pointwise maximization nor standard one-dimensional delegation methods determine the joint optimum. We proceed with the analysis in two steps. First, we fix x_0 and use the Lagrange multiplier approach to verify global optimality of a debt limit (Subsection 3.2). We then choose x_0, prove that the financing constraint binds, and give a curvature condition on the primitives for uniqueness of the necessary conditions for optimality (Subsection 3.3).

3.2 Constrained Debt Limits

Throughout this subsection, we take the initial debt level x_0 as given. Let us define the value function

$$W_0(x_0) = \max_{x_1(\theta) \geq x_0} \mathbb{E}\Big[W_1(x_1, \theta)\Big] \quad \text{subject to (7) and (9).} \tag{11}$$

At each θ where $x_1(\theta)$ is differentiable, the first-order condition of (8) implies the local incentive constraint

$$\frac{\partial V_1\left(x_1(\theta), x_0, \theta\right)}{\partial x_1} \cdot \frac{dx_1(\theta)}{d\theta} = 0. \tag{12}$$

Truthful reporting requires either that debt maximize the borrower's continuation payoff or that it be locally insensitive to the reported type. We call the first case *flexible financing* and define the flexible-debt solution as

$$x_1^f(\theta, x_0) = \arg \max_{x_1 \in [x_0, Y]} V_1(x_1, x_0, \theta) \tag{13}$$

with shorthand $x_1^f(\theta)$ when no confusion arises. It can be verified that the flexible-debt solution is always interior, so it is characterized by the first-order condition:

$$(x_1^f(\theta) - x_0)q'(x_1^f(\theta)) + (1-\theta)q(x_1^f(\theta)) = 0. \tag{14}$$

The next lemma summarizes properties of the flexible-debt solution.

Lemma 2. *The flexible-debt solution $x_1^f(\theta, x_0)$ satisfies the following properties:*

1. *$x_1^f(\theta, x_0)$ is nonincreasing in θ and nondecreasing in x_0.*
2. *$x_1^f(\theta, x_0) \geq x_1^*(\theta)$.*
3. *$x_1^f(\theta, x_0) \geq x_0$. If $\bar{\theta} = 1$, then $x_1^f(\bar{\theta}, x_0) = x_0$.*

Higher θ reduces the benefits of debt issuance, while higher initial debt x_0 strengthens dilution incentives. These incentives explain why flexible borrowing tends to be excessive. Moreover, debt also remains at least x_0, due to the same mechanism as the leverage-ratchet effect (Admati et al., 2018).

To simplify the exposition, we assume:

Assumption 4. $\bar{\theta} = 1$.

Under Assumption 4, the most patient borrower issues no new debt at $t = 1$, so $x_1^f(\bar{\theta}) = x_0$. We can therefore omit debt limits that constrain all types. Since flexible debt is nonincreasing in θ (Lemma 2), any debt limit between x_0 and $x_1^f(\underline{\theta})$ can be represented by a cutoff $\theta_L \in \Theta$ whose flexible debt $x_1^f(\theta_L)$ equals the limit.

We now solve (11) and establish conditions under which the optimal mechanism takes the form of a debt limit

$$x_1^L(\theta) \coloneqq \min\{x_1^f(\theta), x_1^f(\theta_L)\}.$$

This debt policy satisfies the local incentive compatibility conditions (12) and the global condition (9). Figure 2 compares it (solid red) with flexible debt $x_1^f(\theta)$ (dashed red) and efficient debt $x_1^*(\theta)$ (solid green). The debt limit binds at low θ and becomes slack as θ rises, leading to under-borrowing at low θ and over-borrowing at high θ.

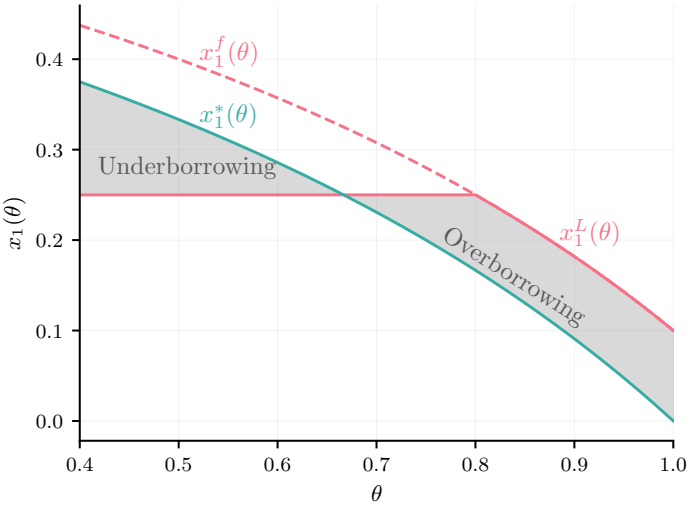


Figure 2: Debt limit and efficient debt level for $\theta \sim U[0.4, 1]$, $Y = 1$, $c(q) = \frac{1}{2}q^2$, $x_0 = 0.1$, and $\theta_L = 0.8$.

Let $W_0^L(x_0)$ denote the value of the following debt-limit problem:

$$\begin{aligned} &\max_{\theta_L \in \Theta} \ \mathbb{E}\left[W_1\left(x_1^f(\theta_L), \theta\right)\mathbf{1}_{\theta \le \theta_L} + W_1\left(x_1^f(\theta), \theta\right)\mathbf{1}_{\theta > \theta_L}\right] \\ &\text{subject to } x_0 \mathbb{E}\left[q\left(\min\{x_1^f(\theta), x_1^f(\theta_L)\}\right)\right] \ge I. \end{aligned} \tag{15}$$

Let λ be the multiplier on the financing constraint and define $\delta = 1/(1+\lambda)$. Since q is decreasing and flexible debt is nonincreasing in θ (Lemma 2), a higher cutoff θ_L tightens the debt limit and raises the expected price of initial debt. The optimal cutoff therefore either maximizes the borrower's expected payoff with slack financing or is determined by a binding financing constraint.

Proposition 1 (Optimal Debt Limit)**.** *Let θ_L^{bc} be the lowest type that satisfies the financing constraint:*

$$\theta_L^{\text{bc}} := \min\left\{\theta_L \in \Theta : \ x_0 \mathbb{E}\left[q\left(\min\{x_1^f(\theta), x_1^f(\theta_L)\}\right)\right] \ge I\right\}.$$

Let θ_L^{uc} be the optimal threshold in the absence of the financing constraint.

- *If* $x_0 > x_1^*\big(\mathbb{E}[\theta]\big)$*, then* $\theta_L^{\text{uc}} = \bar{\theta} = 1$*.*
- *If* $x_0 \leq x_1^*\big(\mathbb{E}[\theta]\big)$*, then* θ_L^{uc} *solves*

$$x_0 \cdot \frac{q'\big(x_1^f(\theta_L)\big)}{q\big(x_1^f(\theta_L)\big)} + \mathbb{E}[\theta_L - \theta \mid \theta \leq \theta_L] = 0. \tag{16}$$

The optimal threshold is $\theta_L = \max\left\{\theta_L^{\text{uc}}, \theta_L^{\text{bc}}\right\}$*, and the corresponding debt limit is* $x_1^L = x_1^f(\theta_L)$*.*

- *If* $\theta_L^{\text{uc}} > \theta_L^{\text{bc}}$*, the financing constraint is slack, and* $\delta = 1$*.*
- *If* $\theta_L^{\text{uc}} \leq \theta_L^{\text{bc}}$*, the financing constraint binds. A corresponding multiplier can be chosen so that*

$$\delta = \min\left\{1,\ -\frac{x_0}{\mathbb{E}[\theta_L^{\text{bc}} - \theta \mid \theta \leq \theta_L^{\text{bc}}]} \frac{q'\big(x_1^f(\theta_L^{\text{bc}})\big)}{q\big(x_1^f(\theta_L^{\text{bc}})\big)}\right\}. \tag{17}$$

The first part of the proposition states the solution when the financing constraint is slack, the optimality condition in equation (16) balances the welfare loss from constraining low-θ borrowers against excessive borrowing by high-θ borrowers. The second part considers the case when the financing constraint binds and a tighter debt limit is required.

Proposition 1 considers the optimization within the class of debt limits. The next proposition establishes when this candidate is globally optimal among all incentive-compatible debt schedules for a given x_0. The complete analysis of the general problem is developed in the Supplemental Appendix.

Proposition 2 (Sufficient Conditions for Optimality)**.** *Consider the debt limit in Proposition 1. Let* $\delta = 1/(1+\lambda)$ *where* λ *is the Lagrange multiplier of the financing constraint in* (15)*. If the function*

$$\delta G(\theta) + \frac{x_0 q'(x_1^f(\theta))}{q(x_1^f(\theta))} g(\theta) \tag{18}$$

is non-decreasing for $\theta \in [\theta_L, \bar{\theta}]$*. Then,* $x_1^{n*}(\theta) = \min\{x_1^f(\theta), x_1^f(\theta_L)\}$ *solves the mechanism design problem* (11)*.*

Following the Lagrangian approach of Amador, Werning, and Angeletos (2006) and Amador and Bagwell (2013), we construct, for a given initial debt, a multiplier on the incentive constraints that makes the proposed debt limit maximize the Lagrangian. Above the cutoff, the multiplier makes the marginal effect of changing debt zero. Below the cutoff, the optimal-cutoff condition and monotonicity of debt rule out an improving deviation. The verification condition in Proposition 2 ensures that the multiplier is nondecreasing; together with the curvature assumptions, this makes the Lagrangian concave. Feasibility and complementary slackness then establish global optimality. The complete proof is in Supplemental Appendix OA.1.1.

In the linear example, we can provide a sufficient condition on the density of types that ensures that condition (18) holds.

Linear Example. *Under non-exclusive lending,*

$$x_1^f(\theta, x_0) = \frac{(1-\theta)Y + x_0}{2-\theta}, \qquad q(x_1^f(\theta, x_0)) = \frac{Y - x_0}{c}\frac{1}{2-\theta}.$$

A sufficient condition for (18) *is*

$$\sup_{\theta\in\Theta}(2-\theta)\frac{g'(\theta)}{g(\theta)} \leq 1.$$

In what follows, we restrict attention to debt limits and consider the optimal debt limit with value function $W_0^L(x_0)$. If condition (18) holds for every feasible x_0, then $W_0^L(x_0) = W_0(x_0)$ throughout the feasible set, and the restriction to debt limits is without loss of generality. When this is not the case, our results characterize the optimal debt limit, which need not solve the unrestricted mechanism-design problem.

3.3 Initial Debt and Implementation

We now proceed to the second step in the analysis and find the optimal value of x_0 by solving the problem:

$$\begin{aligned} &\max_{x_0} \; W_0^L(x_0) \\ &\text{subject to } x_0 q(x_0) \geq I, \end{aligned} \tag{19}$$

where $W_0^L(x_0)$ is the value of (15). The constraint $x_0 q(x_0) \geq I$ is both sufficient and necessary for the feasible set in problem (15) to be non-empty. This constraint holds if and only if $x_0 \in [x_0^{\min}, x_0^{\max}]$, where $x_0^{\min} \leq x_0^{\max}$ are the lower and upper solutions to $x_0 q(x_0) = I$, which exist given $I < I^{\max}$ (Assumpion 2).

Figure 3 illustrates problem (19). The financing constraint requires $\theta_L \geq \theta_L^{\text{bc}}(x_0)$, where $\theta_L^{\text{bc}}(x_0)$ is the lowest cutoff that allows initial debt x_0 to raise I. For a given x_0, the borrower chooses θ_L to balance the costs of restricting financing against the costs of dilution. For a given θ_L, higher initial debt x_0 lowers the borrower's payoff by strengthening the incentive to dilute existing lenders. The indifference curves depict this tradeoff, with darker curves representing higher payoffs. When the optimum retains financial flexibility, the borrower chooses the tangency between an indifference curve and $\theta_L^{\text{bc}}(x_0)$.

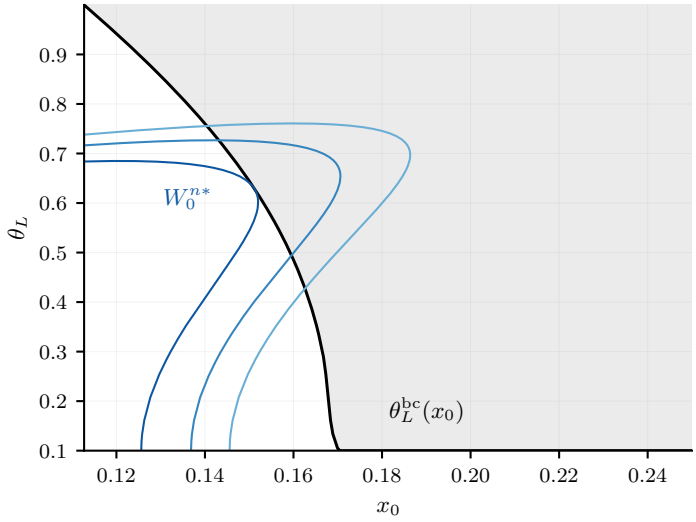


Figure 3: Indifference curves and optimal pair (x_0, θ_L) for $\theta \sim U[0.1, 1]$, $Y = 1$, $c(q) = \frac{1}{2}q^2$, and $I = 0.4 \times I^{\max}$.

The next proposition characterizes the optimal initial borrowing and its associated debt limit.

Proposition 3 (Optimal initial debt: necessary conditions)**.** *Let x_0^{n*} solve (19), with optimal cutoff θ_L^{n*}. Define*

$$x_f^{n*}(\theta) := x_1^f(\theta, x_0^{n*}), \qquad x_L^{n*} := x_f^{n*}(\theta_L^{n*}).$$

1. *No additional borrowing: If $x_1^*(\mathbb{E}[\theta]) \leq x_0^{\min}$, then $x_0^{n*} = x_0^{\min}$ and $\theta_L^{n*} = 1$.*

2. *Financial flexibility: If* $x_1^*(\mathbb{E}[\theta]) > x_0^{\min}$*, then* $x_0^{n*} > x_0^{\min}$*,* $\theta_L^{n*} < 1$*, and* x_0^{n*}, θ_L^{n*} *jointly solve:*

$$\begin{aligned} x_0^{n*}\mathbb{E}\left[q(x_L^{n*})\mathbf{1}_{\{\theta\le\theta_L^{n*}\}} + q(x_f^{n*}(\theta))\mathbf{1}_{\{\theta>\theta_L^{n*}\}}\right] &= I \\ x_0^{n*} &= \frac{(1-\delta)\mathbb{E}\left[q(x_L^{n*})\mathbf{1}_{\{\theta\le\theta_L^{n*}\}} + q(x_f^{n*}(\theta))\mathbf{1}_{\{\theta>\theta_L^{n*}\}}\right]}{-\mathbb{E}\left[q'(x_f^{n*}(\theta))\frac{\partial x_1^f(\theta,x_0^{n*})}{\partial x_0}\mathbf{1}_{\{\theta>\theta_L^{n*}\}}\right]} \end{aligned} \tag{20}$$

where δ *is given by* (17)*, evaluated at* $x_0 = x_0^{n*}$ *and* $\theta_L^{\text{bc}} = \theta_L^{n*}$*.*

When $x_1^*(\mathbb{E}[\theta]) \le x_0^{\min}$, the initial financing requirement already forces debt to at least the ex-ante optimal level. Dilution incentives would encourage even further borrowing, so the optimal debt limit rules out additional issuance. When $x_1^*(\mathbb{E}[\theta]) > x_0^{\min}$, the optimal policy preserves some flexibility to balance over- and under-borrowing. Borrowers with low liquidity needs (high θ) can then borrow flexibly at $t = 1$.

Notice that Proposition 3 also implies that the financing constraint always binds: the borrower chooses initial debt and the debt limit to raise exactly I at $t = 0$. Intuitively, any additional initial borrowing would strengthen dilution incentives at $t = 1$, reducing future borrowing flexibility.

In the linear example, we can also provide a simple condition on the financing needs that ensures that the borrower retains some financial flexibility.

Linear Example. *With quadratic effort costs, debt capacity is* $I^{\max} = Y^2/(4c)$*. Define the financing-needs parameter*

$$\kappa := cI/Y^2 = I/(4I^{\max}),$$

which measures investment relative to debt capacity. Proposition 3 implies that preserving financial flexibility requires

$$\mathbb{E}[\theta] < \frac{2\sqrt{1-4\kappa}}{1+\sqrt{1-4\kappa}}.$$

To implement the optimal allocation, the borrower issues long-term debt with face value x_0^{n*} at $t = 0$ and includes an incurrence covenant limiting total debt due at $t = 2$ to x_L^{n*}. The borrower can obtain financing from new lenders at $t = 1$ provided total debt stays within this limit. The covenant protects initial lenders without requiring exclusive lending.

Practitioners emphasize the trade-off between creditor protection and issuer flexibility in covenant design. Simpson Thacher & Bartlett LLP (2022, p. 2) describe high-yield covenants as balancing creditor protection against the issuer's flexibility to operate and grow. In our model, the debt limit captures this trade-off by restricting dilution while allowing further borrowing.

The next proposition gives a condition on inverse-demand elasticity that makes the objective single-peaked in initial debt. Under this condition, the necessary conditions in Proposition 3 identify a unique optimum whenever some financial flexibility remains. Relaxing the debt limit is valuable when flexibility is scarce, but the added dilution cost eventually outweighs the benefit.

Proposition 4 (Sufficient Conditions Initial Debt)**.** *Define the elasticity of the inverse demand for debt by*

$$\varepsilon(x) := -\frac{xq'(x)}{q(x)}$$

and define

$$\Gamma(x) := \frac{x\varepsilon'(x) - \varepsilon(x)}{\varepsilon(x)^2}.$$

If Γ *is nonincreasing on the relevant debt interval, then the necessary conditions in Proposition 3 are also sufficient. In particular, if* $x_1^*(\mathbb{E}[\theta]) > x_0^{\min}$*, then the system in* (20) *has a unique solution, and this solution is the unique maximizer of* (19)*.*

The condition in Proposition 4 is satisfied for many common cost functions, including the quadratic cost function in the linear case. For example, when $c(q) = cq^\eta$ with $\eta \geq 2$, $\Gamma(x) = \eta - 1$ is constant, satisfying the condition in Proposition 4.

3.4 Comparative Statics

Next, we examine how the investment requirement I affects the optimal contract when some financial flexibility remains, $\theta_L^{n*} < 1$.

Proposition 5 (Comparative statics with respect to investment)**.** *Suppose that the condition on* Γ *in Proposition 4 holds. For investment levels satisfying* $x_1^*(\mathbb{E}[\theta]) > x_0^{\min}$*,* x_0^{n*} *and* θ_L^{n*} *are strictly increasing in* I*, while* x_L^{n*} *is strictly decreasing in* I*.*

A larger initial claim strengthens the incentive to dilute existing lenders, so the borrower commits to a lower limit on total debt. The room for additional debt issuance, therefore, shrinks both because the initial claim is larger and because the debt

limit is lower. The borrower is constrained by the limit for a wider range of liquidity shocks, and the lower debt limit increases under-borrowing relative to the efficient benchmark when liquidity needs are severe. Borrowers who remain unconstrained instead choose higher total debt, increasing over-borrowing.

Figure 4 illustrates the effect of higher financing needs κ: the debt limit falls and binds for more types, while types that remain unconstrained choose higher total debt.

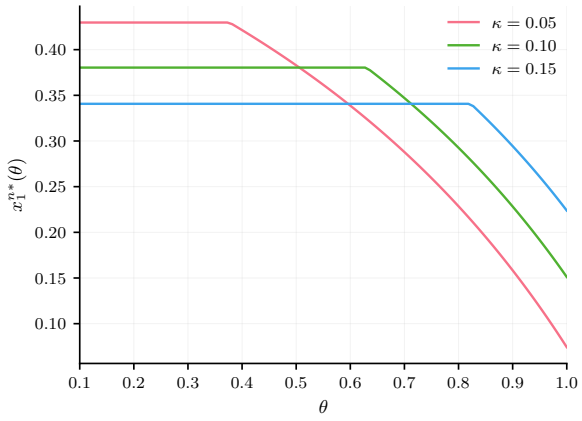


Figure 4: Comparative statics of face value at $t = 1$ for changes in financing need κ, with $\theta \sim U[0.1, 1]$, $Y = 1$, and $c(q) = \frac{1}{2}q^2$.

We also brieefly explore the impact of changes in the distribution of liquidity shocks. In particular, we consider changes in the density g that shift toward higher θ, and show that they can either increase or reduce financial flexibility. On the one hand, the shift reduces the borrower's need for additional financing and hence its willingness to pay for flexibility. On the other hand, it also reduces existing lenders' exposure to dilution, which can lower the cost of allowing future borrowing. The lower financing need favors less flexibility, while the lower cost favors more. The effect depends on which force dominates. Figure 5 illustrates both possibilities: a shift toward higher θ raises the optimal cutoff when financing needs are high (left panel) and lowers it when financing needs are low (right panel).

4 Exclusive Lending

We now turn to consider the case of exclusive lending, under which the borrower commits to return to a single lender for refinancing. Because the lender internalizes

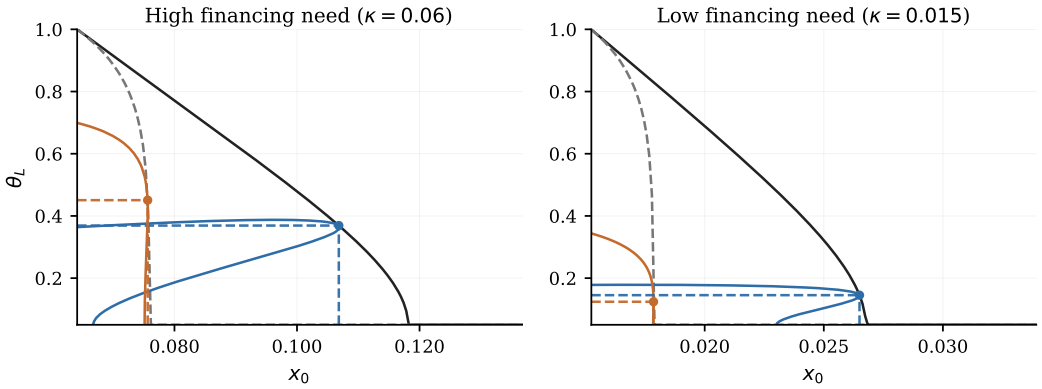


Figure 5: Distribution shifts and the optimal debt-limit cutoff. Financing schedules (black and gray) and indifference curves (blue and orange) correspond to $\gamma = -5$ and 5, respectively. We use $g(\theta;\gamma) \propto e^{\gamma\theta}$ on $[0.05, 1]$, $Y = 1$, and $c(q) = \frac{1}{2}q^2$, with κ indicated above each panel.

the effect of new borrowing on its existing claim, this allocation provides a benchmark for the contracts considered before.

As before, the borrower privately observes the discount-factor shock θ before refinancing. By the revelation principle, we can restrict attention to direct mechanisms specifying an initial transfer τ_0, a refinancing transfer $\tau_1(\theta)$, and total debt $x_1(\theta)$. The borrower's expected payoff is

$$\tau_0 - I + \mathbb{E}\left[\tau_1(\theta) + \theta V_2(x_1(\theta))\right].$$

The lender's participation constraint is $\mathbb{E}[x_1(\theta)q(x_1(\theta)) - \tau_1(\theta)] \geq \tau_0$. Truthful reporting requires

$$\theta \in \arg\max_{\hat{\theta}} \ \tau_1(\hat{\theta}) + \theta V_2(x_1(\hat{\theta})). \tag{21}$$

Using the envelope theorem, we can rewrite (21) as

$$\begin{aligned} &\tau_1(\theta) + \theta V_2\left(x_1(\theta)\right) = \tau_1(\bar{\theta}) + \bar{\theta} V_2(x_1(\bar{\theta})) - \int_{\theta}^{\bar{\theta}} V_2(x_1(\tilde{\theta}))d\tilde{\theta}, \ \forall \theta \in \Theta \\ &x_1(\theta) \text{ is non-increasing.} \end{aligned} \tag{22}$$

The lender's participation constraint binds. Substituting the envelope condition (22)

into it gives the initial financing constraint:

$$\tau_0 = \mathbb{E}\left[W_1(x_1(\theta),\theta) + \frac{G(\theta)}{g(\theta)} V_2\left(x_1(\theta)\right)\right] - \tau_1(\bar{\theta}) - \bar{\theta} V_2(x_1(\bar{\theta})) \geq I. \tag{23}$$

Substituting τ_0 and $\tau_1(\theta)$ into the objective function and omitting the slack constraint $x_1(\theta) \in [0, Y]$, the optimal contract solves

$$\begin{aligned} &\max_{\{x_1(\theta),\tau_1(\theta)\geq 0\}} \mathbb{E}\left[W_1(x_1(\theta),\theta)\right] - I \\ &\text{subject to (22) and (23)} \end{aligned} \tag{24}$$

We make the following assumption to ensure that the previous optimization problem is convex:

Assumption 5. *For all $\theta \in \Theta$, $W_1(x_1,\theta) + [G(\theta)/g(\theta)]V_2(x_1)$ is concave in x_1. Under Assumption 1, this holds if*

$$2 + \frac{x_1 q''(x_1)}{q'(x_1)} \geq \bar{\theta} + \frac{1}{g(\bar{\theta})}.$$

Linear Example. *In the linear example with quadratic cost of effort, under the assumption that $G(\theta)$ is log-concave, Assumption 5 holds whenever $(2-\bar{\theta})g(\bar{\theta}) \geq 1$.*

It can be verfied that the transfer schedule in equation (21) is nonincreasing in θ (Supplemental Appendix Lemma OA.1), reflecting the greater borrowing needs of low-θ types. Therefore, the constraint $\tau_1(\theta) \geq 0$ only need to be imposed on type $\bar{\theta}$. Moreover, it is optimal to set $\tau_1(\bar{\theta}) = 0$: if it were positive, the borrower could reduce all transfers by the same amount and raise more funds at $t = 0$, relaxing the financing constraint.[5]

We solve the problem in two steps. First, let λ denote the Lagrange multiplier on the financing constraint $\tau_0 \geq I$, and define $\delta := 1/(1+\lambda)$. For a given δ, define the virtual type

$$\phi_\delta(\theta) := \theta + (1-\delta)\frac{G(\theta)}{g(\theta)}. \tag{25}$$

[5]With no discounting between $t = 0$ and $t = 1$, the borrower is indifferent when the financing constraint is slack; $\tau_1(\bar{\theta}) = 0$ then requires a slight preference for borrowing at $t = 0$.

The distortion from incentive provision is captured by evaluating payoffs at the virtual type, since

$$W_1(x_1, \phi_\delta(\theta)) = W_1(x_1, \theta) + (1-\delta)\frac{G(\theta)}{g(\theta)}V_2(x_1).$$

Given this, the optimal contract solves

$$\begin{aligned} &\max_{\{x_1(\theta)\}} \ \mathbb{E}\left[W_1(x_1(\theta), \phi_\delta(\theta))\right] - (1-\delta)\bar{\theta}V_2(x_1(\bar{\theta})) \\ &\text{subject to } x_1(\theta) \text{ is non-increasing.} \end{aligned} \tag{26}$$

The pointwise optimum is $x_1^*(\phi_\delta(\theta))$. It is nonincreasing under Assumption 1 and coincides with the efficient rule $x_1^*(\theta)$ when $\delta = 1$. For $\delta < 1$, however, the final term in (26) prevents pointwise maximization from solving the problem, so we need to explicitly incorporate the monotonicity constraint.[6]

Proposition 6. *For a given δ, let $x_1^{s*}(\theta)$ denote the optimal debt schedule solving problem (26).*

- *If $\delta \geq \underline{\theta}/\mathbb{E}[\theta]$, then*

$$x_1^{s*}(\theta) = \max\{x_1^*(\phi_\delta(\theta)), x_1^*(\phi_\delta(\theta_H))\},$$

  where $\theta_H \in \Theta$ solves:

$$\begin{aligned} &\int_{\theta_H}^{\bar{\theta}} \frac{\partial W_1(x_1^*(\phi_\delta(\theta_H)), \theta)}{\partial x_1} dG(\theta) \\ &\quad + (1-\delta)q\left(x_1^*(\phi_\delta(\theta_H))\right)\left(\theta_H G(\theta_H) + \int_{\theta_H}^{\bar{\theta}} \theta dG(\theta)\right) = 0. \end{aligned} \tag{27}$$

- *If $\delta < \underline{\theta}/\mathbb{E}[\theta]$, $x_1^{s*}(\theta) = x_1^*(\delta\mathbb{E}[\theta])$ for all $\theta \in \Theta$.*

The optimal transfers $\{\tau_0, \tau_1(\theta)\}$ satisfy (22) and (23).

On $[\underline{\theta}, \theta_H)$, the incentive cost $(1-\delta)G(\theta)/g(\theta)$ makes the principal choose debt as if the borrower had a higher discount factor, reducing debt relative to the efficient

[6] The final term involves only $x_1(\bar{\theta})$, and since $V_2' = -q$ it raises the marginal benefit of debt for the highest type alone. Unless $\delta = 1$, pointwise maximization yields a schedule that jumps up at $\bar{\theta}$ and violates monotonicity.

allocation. Borrowers with higher θ need less financing at $t = 1$ and receive less debt and lower transfers. To raise more funds at $t = 0$, the principal would like to reduce transfers further and pledge more cash flow. Once transfers reach zero, limited liability prevents further reductions, and incentive compatibility requires constant debt above θ_H. Thus, whenever the financing constraint binds, high-θ types pool because the borrower seeks to pledge more of the project's cash flow at $t = 0$ by reducing their $t = 1$ transfers until limited liability binds. As in Castro-Pires and Moreira (2021), binding limited liability limits the use of transfers to separate types, leading to pooling.

The previous analysis provides the solution for a given multiplier δ. The next result determines the value of the multiplier by considering the initial financing constraint.

Proposition 7. *Let*

$$I^s := \mathbb{E}\left[W_1\left(x_1^*(\theta), \theta\right) + V_2\left(x_1^*(\theta)\right)\frac{G(\theta)}{g(\theta)}\right] - \bar{\theta} V_2\left(x_1^*(\bar{\theta})\right)$$

and

$$I^p := x_1^*(\underline{\theta}) q\left(x_1^*(\underline{\theta})\right),$$

where $x_1^*(\theta)$ is defined in (5). The optimal debt schedule is given by Proposition 6 at $\delta = \delta^{s*}$, determined as follows:

- *If $I \leq I^s$, the optimal policy is $x_1^{s*}(\theta) = x_1^*(\theta)$, and $\delta^{s*} = 1$.*
- *If $I^s < I < I^p$, the schedule is partially separating. The values $\delta^{s*} \in (\underline{\theta}/\mathbb{E}[\theta], 1)$ and $\theta_H^{s*} \in (\underline{\theta}, \bar{\theta})$ jointly solve (27) and $\tau_0 = I$.*
- *If $I^p \leq I \leq I^{\max}$, the schedule is fully pooled, and $\delta^{s*} \in [0, \underline{\theta}/\mathbb{E}[\theta]]$ solves*

$$x_1^*\left(\delta^{s*}\mathbb{E}[\theta]\right) q\left(x_1^*\left(\delta^{s*}\mathbb{E}[\theta]\right)\right) = I.$$

The threshold I^s is the expected pledgeable income under the efficient debt schedule, net of the interim transfers required for truthful reporting. Relative to a contractible shock, these transfers reduce initial lending. Information rents do not distort debt when financing is slack (Baron and Besanko, 1984). In that case, interim transfers can provide the rents required for truthful reporting without affecting initial financing, so the efficient debt schedule remains feasible. Under Assumption 4, $I^s = 0$,

so financing binds for every positive investment. When financing binds, information rents absorb scarce pledgeable income, biasing the optimal schedule toward higher debt; consequently, the no-distortion result of Baron and Besanko (1984) no longer applies. At $I \geq I^p$, all types pool with zero interim transfers, and the common debt level rises with financing needs.

Our next result compares the exclusive lending allocation to the optimal simple debt contract under non-exclusive lending.

Proposition 8. *Suppose that Assumption 4 holds and that* $\theta_L^{n*}, \theta_H^{s*} \in (\underline{\theta}, \bar{\theta})$*, where the two thresholds are defined in Propositions 3 and 7. There exists a unique* $\theta^\dagger \in (\underline{\theta}, \min\{\theta_H^{s*}, \theta_L^{n*}\})$ *such that* $x_1^{n*}(\theta^\dagger) = x_1^{s*}(\theta^\dagger)$*, and*

$$x_1^{n*}(\theta) < x_1^{s*}(\theta) \quad \text{for } \theta < \theta^\dagger, \qquad x_1^{n*}(\theta) > x_1^{s*}(\theta) \quad \text{for } \theta > \theta^\dagger.$$

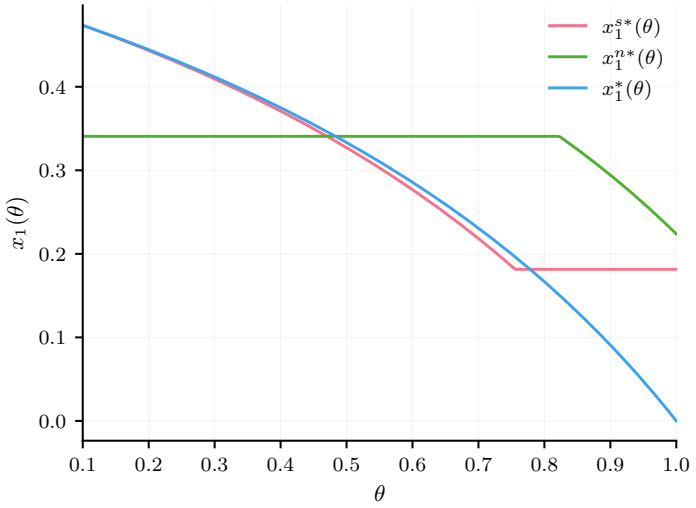


Figure 6: Exclusive and non-exclusive lending. The figure plots total debt at $t = 1$ under exclusive lending, $x_1^{s*}(\theta)$, non-exclusive lending, $x_1^{n*}(\theta)$, and complete information, $x_1^*(\theta)$. We assume $\theta \sim U[0.1, 1]$, $Y = 1$, $c(q) = \frac{1}{2}q^2$, and $I = 0.6 \times I^{\max}$.

Figure 6 illustrates the comparison in Proposition 8. Under non-exclusive lending, the debt limit leads to underborrowing after severe shocks and overborrowing after mild shocks. Because of incentive compatibility, even some borrowers with severe shocks who hit the debt limit overborrow relative to the exclusive-lending benchmark. It is noteworthy that the pooling regions also differ across the two regimes: borrowers

with severe shocks pool under non-exclusive lending, whereas borrowers with mild shocks pool under exclusive lending.

Implementation. The exclusive-lending allocation can be implemented as a long-term contract signed at $t = 0$ together with a state-contingent credit line at $t = 1$. Let $x_0^s \coloneqq x_1^{s*}(\bar{\theta})$ denote the face value of outstanding debt for the type with the lowest refinancing need. If the solution is fully pooled, then $x_1^{s*}(\theta) = x_0^s$ for all θ, so there is no refinancing margin at $t = 1$. Otherwise, each type on $[\underline{\theta}, \theta_H]$ chooses additional debt

$$\Delta x_1^s(\theta) \coloneqq x_1^{s*}(\theta) - x_0^s.$$

The next proposition shows how the exclusive-lending allocation can be implemented with a nonlinear price schedule for new debt.

Proposition 9 (Credit-line implementation)**.** *Suppose $\theta_H > \underline{\theta}$, and let θ^s denote the inverse of Δx_1^s on $[\underline{\theta}, \theta_H]$. The nonlinear price schedule*

$$p_1^s(\Delta x_1) \coloneqq \frac{1}{\Delta x_1} \int_0^{\Delta x_1} \theta^s(z) q(x_0^s + z)\, dz, \quad \tau_1^{s*}(\theta) = \Delta x_1^s(\theta)\, p_1^s(\Delta x_1^s(\theta)), \qquad \theta \in [\underline{\theta}, \theta_H]$$

implements the exclusive allocation. For every positive draw, $p_1^s(\Delta x_1) < q(x_0^s + \Delta x_1)$.

Note that the credit-line price $p_1^s(\Delta x_1)$ is below the competitive spot price $q(x_0^s + \Delta x_1)$, so the borrower pays a higher implicit interest rate than what he would pay in a competitive market. Hence, the contract can only be implemented if the borrower cannot access outside financing, since the borrower would otherwise prefer to issue the same debt at the higher market price. Moreover, the existing lender must also commit to provide state-contingent lending at $t = 1$, which may be difficult when lenders are dispersed or face liquidity shortages.

5 Richer Contracting Mechanism

We now consider two contractual provisions that compensate existing lenders for dilution when the borrower issues new debt. Performance-sensitive debt adjusts the amount owed to existing lenders at $t = 2$, whereas contingent prepayment uses part of the new borrowing proceeds at $t = 1$ to repay existing debt. Both provisions make the terms of existing debt contingent on subsequent contracts. By making the

borrower internalize dilution, performance-sensitive debt and contingent prepayment, can implement the exclusive-lending allocation.

5.1 Performance-Sensitive Debt

In Section 3, we restricted attention to fixed-rate debt contracts whose face value x_0 does not adjust to subsequent borrowing. An alternative is to let the interest rate adjust to changes in the borrower's credit risk. Contracts with this feature can be thought of as performance-sensitive debt.

In loan agreements, this adjustment typically takes the form of pricing grids that link the loan spread to credit ratings, debt-to-EBITDA ratios, or coverage ratios. For example, Asquith, Beatty, and Weber (2005) document that performance-pricing grids appear in 41% of loans and account for 54% of loan volume in their sample of 8,761 bank loans issued over 1995–1998. They argue that specifying rate reductions in advance reduces the need for costly renegotiation, particularly in syndicated loans. Our analysis points to a complementary role: under non-exclusive lending, performance-sensitive terms can preserve future financing while compensating existing lenders for dilution.

We now show how a performance-sensitive debt contract can implement the exclusive allocation. Let $\{\tau_0^{s*}, \tau_1^{s*}(\theta), x_1^{s*}(\theta)\}$ denote the exclusive-lending allocation characterized in Section 4, and let $x_0^s := x_1^{s*}(\bar{\theta})$ denote the baseline existing debt claim. For each report θ, define the existing debt claim

$$x_0^{\mathrm{ps}}(\theta) := x_1^{s*}(\theta) - \frac{\tau_1^{s*}(\theta)}{q(x_1^{s*}(\theta))}$$

and the new debt issued at $t = 1$ by

$$\Delta x_1^{\mathrm{ps}}(\theta) := x_1^{s*}(\theta) - x_0^{\mathrm{ps}}(\theta) = \frac{\tau_1^{s*}(\theta)}{q(x_1^{s*}(\theta))}.$$

Following the standard logic of nonlinear pricing, we make the existing debt claim contingent on observed new issuance to implement each type's allocation. The next proposition shows that performance-sensitive debt can be used to reproduce the exclusive-lending allocation for every issuance level on the equilibrium path.[7]

[7]The initial contract limits aggregate new issuance to the largest issue prescribed

Proposition 10. *Suppose the exclusive-lending allocation is not fully pooled, so* $\theta_H^{ss} > \underline{\theta}$. *On* $[\underline{\theta}, \theta_H^{ss}]$, *the issuance schedule* $\Delta x_1^{\text{ps}}(\theta)$ *is strictly decreasing. Let* $\theta^{\text{ps}}(\Delta x_1)$ *denote the type prescribed to issue* Δx_1. *The existing-claim schedule*

$$x_0^{\text{ps}}(\Delta x_1) = x_1^{ss}(\theta^{\text{ps}}(\Delta x_1)) - \Delta x_1$$

reproduces the allocation under exclusive lending.

Under the credit-line implementation, the initial lender limits dilution by providing less cash per dollar of additional debt than competitive outside lenders would provide. Under performance-sensitive debt, new lenders provide financing on competitive terms, while the existing debt claim adjusts to compensate the initial lender for dilution. The corresponding adjustment leads to the same allocation. Figure 7(a) shows the existing claim increasing with new issuance in the linear example.

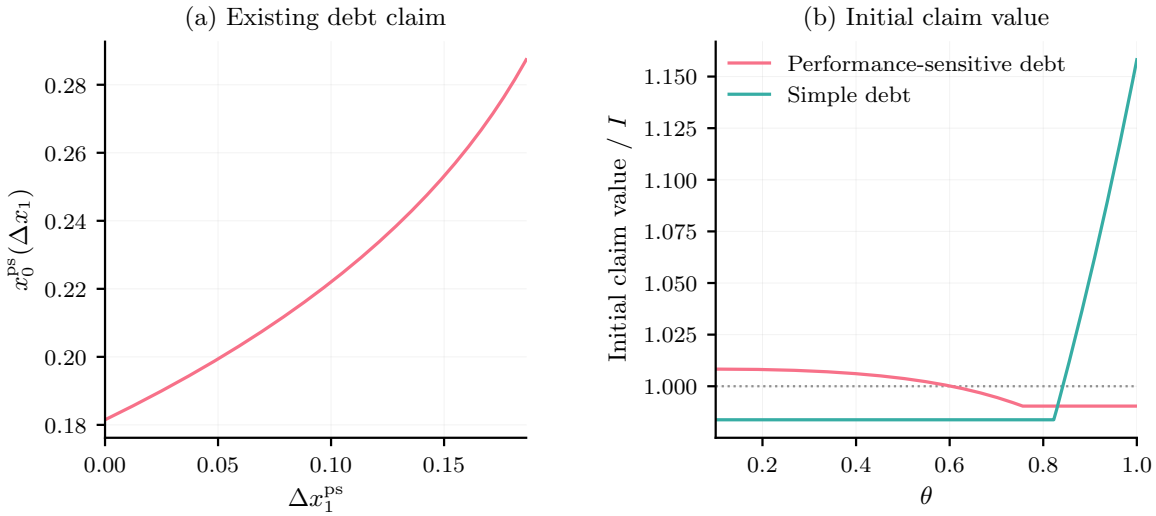


Figure 7: Performance-sensitive debt. Panel (a) plots the existing claim against new issuance. Panel (b) compares the initial claim's expected repayment conditional on θ, normalized by I, under performance-sensitive debt and the optimal debt limit. We assume $\theta \sim U[0.1, 1]$, $Y = 1$, $c(q) = \frac{1}{2}q^2$, and $\kappa = 0.15$.

We next compare dilution under performance-sensitive debt and simple debt with an optimal debt limit. Figure 7(b) plots the $t = 1$ value of the initial claim under

by the schedule.

each contract. Under performance-sensitive debt, this value is

$$L(\theta) := x_0^{\text{ps}}(\theta) q(x_1^{s*}(\theta)) = x_1^{s*}(\theta) q(x_1^{s*}(\theta)) - \tau_1^{s*}(\theta).$$

On $[\underline{\theta}, \theta_H^{s*})$, the envelope and optimality conditions imply (Supplemental Appendix Lemma OA.2)

$$L'(\theta) = (1 - \delta^{s*}) \frac{G(\theta)}{g(\theta)} q(x_1^{s*}(\theta)) x_1^{s*\prime}(\theta) \leq 0.$$

Performance-sensitive debt protects initial lenders against dilution by increasing the face value of their claims when new debt is issued. An adjustment that merely offsets dilution would leave the value of those claims, $L(\theta)$, constant across states. When the initial financing constraint binds, the optimal contract also shifts repayment toward low-θ states, where the borrower values future cash flow less. This allocation of repayment, subject to incentive compatibility, makes $L(\theta)$ decrease with θ on $[\underline{\theta}, \theta_H^{s*})$. When the financing constraint is slack, $\delta^{s*} = 1$, the contract exactly offsets dilution and $L(\theta)$ is constant.

Our model offers a new rationale for performance-sensitive debt. Manso, Strulovici, and Tchistyi (2010) show how such debt can screen high-growth borrowers from low-growth borrowers under asymmetric information, while Bensoussan, Chevalier-Roignant, and Rivera (2021) study how performance sensitivity can mitigate debt overhang by changing shareholders' payoffs from exercising growth options. Chaigneau, Edmans, and Gottlieb (2021) show that performance-sensitive debt can arise in the presence of moral hazard when additional performance signals are contractible. Diamond, Hu, and Rajan (2022) show that, under aggregate uncertainty, rate increases following covenant violations can expand borrowing capacity by strengthening borrowers' incentives to improve cash-flow pledgeability. In our model, performance sensitivity instead prices the externality from non-exclusive borrowing: new issuance triggers an adjustment to the existing claim that compensates initial lenders for dilution.

Implementing contracts that compensate existing lenders for dilution can be challenging when lenders are dispersed. These contracts require lenders to track total leverage, claim priority, liens, guarantees, leases, and other debt-like liabilities, including obligations not classified as debt in financial statements. Designing the adjustment schedule adds another challenge: the contract must specify how existing claims change at each level of new issuance, accounting for the distribution of future

financing shocks and the effect of additional borrowing on their value. By contrast, even though a debt limit still requires monitoring, it sets a single cutoff and requires less information about the shock distribution. This simpler design may make debt limits a more practical alternative when lenders are dispersed.

5.2 Contingent Prepayment

State-contingent debt acceleration can also implement the exclusive-lending allocation. In practice, a debt sweep requires the borrower to use part of the proceeds from new debt to prepay existing lenders. It compensates lenders at $t = 1$, whereas performance-sensitive debt adjusts their $t = 2$ claim. Under a standard restrictive covenant, new issuance can trigger the acceleration of the entire loan; a sweep makes prepayment conditional on how much the borrower issues. Prepayment need not absorb all issuance proceeds, so the borrower can still obtain new liquidity.

Loan agreements use debt sweeps as part of a broader class of mandatory-prepayment provisions.[8] They appear in private corporate loans, leveraged loans, and debtor-in-possession facilities (Bradley and Roberts, 2015; Eckbo, Li, and Wang, 2023). Sweeps became prominent in bank loan contracts during the 1990s (Bratton, 2016). Between 1993 and 2001 the share of loans with a debt sweep rose from 18% to 80% (Bradley and Roberts, 2015), though looser covenants in more recent years have reduced their use (Bratton, 2016). Eckbo et al. (2023) finds debt-issuance sweeps in 26.2% of debtor-in-possession loan packages and 26.9% of matched leveraged-loan packages, compared with prepayment-sweep rates of 51.3% and 35.8%, respectively.[9]

We can implement the exclusive benchmark with a nonlinear prepayment schedule that gives the borrower the same net cash and total debt for each allowed issue, so no type gains by choosing the issue meant for another. The prepayment share must vary with the size of the issue: larger issues come from lower-θ states and dilute the existing claim more, so a nonlinear schedule makes the borrower internalize that dilution while preserving access to outside debt. The contract cannot require prepayments above existing debt as it can only accelerate outstanding claims. This restriction holds

[8] Equity and asset sweeps direct a share of equity-issuance and asset-sale proceeds, respectively, to debt repayment.

[9] Avon's 2012 loan required prepayment of 50% of net cash proceeds from debt issuance exceeding $500 million (Avon Products, Inc., 2014).

whenever the exclusive lender's net repayment in the state with the greatest liquidity need does not exceed the initial claim, $x_1^{s*}(\underline{\theta})q(x_1^{s*}(\underline{\theta})) - \tau_1^{s*}(\underline{\theta}) \leq x_0^s$. Supplemental Appendix OA.4.2 states the prepayment schedule and proves that it replicates the exclusive-lending allocation under this condition.

5.3 Short-Term Debt

Our previous work shows that short-term debt can resolve the dilution problem (Hu and Varas, 2025; Hu, Varas, and Ying, 2026). In this setting, if the borrower can use both simple long-term debt and short-term debt, it prefers to finance exclusively with short-term debt. At refinancing, the borrower chooses the ex-post efficient debt level whenever it raises enough funds to repay the maturing debt and otherwise borrows just enough to meet that obligation. This allocation dominates what can be achieved with simple long-term debt subject to a debt limit and is therefore preferred by the borrower.

However, short maturity alone cannot reproduce the exclusive-lending allocation. Because higher debt weakens effort incentives, additional financing entails a debt-overhang cost. The optimal exclusive-lending allocation concentrates this inefficiency in low-θ states, where the benefit of additional financing is greatest. Implementing this allocation requires cross-subsidization across states, which short-term debt alone cannot provide. Moreover, our comparison of short- and long-term debt abstracts from rollover transaction costs and uncertainty about intermediate cash flows. In a related setting with such uncertainty, Hu et al. (2026) show that borrowers issue long-term debt when short-term debt becomes risky.

6 Applications Beyond Corporate Debt

Our delegation approach to non-exclusive contracting can be applied to other settings with a similar incentive problems, such as sovereign borrowing and intertemporal product markets. The initial contract creates claims that later actions can dilute, so the contract must protect these claims while leaving discretion to respond to private information.

6.1 Sovereign Borrowing

For a sovereign, θ may represent privately observed spending needs. Borrowing to meet these needs dilutes outstanding bonds, a loss the sovereign does not internalize. This commitment problem can dissipate the gains from borrowing when governments issue almost continuously (DeMarzo, He, and Tourre, 2023). Fiscal rules that limit debt, deficits, or spending (Alonso et al., 2025) protect bondholders, but also restrict the sovereign's response to spending needs.[10] Our model makes this tradeoff depend on the initial debt chosen alongside the borrowing restriction. More initial debt strengthens dilution incentives and makes future borrowing discretion more costly. Under the conditions of Proposition 5, greater financing needs therefore raise initial debt and tighten the optimal debt limit.

The sovereign can also protect bondholders by compensating them when it borrows more. In Hatchondo, Martinez, and Sosa-Padilla (2016), a covenant offsets the price decline caused by new borrowing, holding the state fixed. Our compensation schedule must also induce truthful reporting of liquidity needs. This requirement matters when initial financing binds: the contract shifts pledgeable income toward states with greater liquidity needs, where the sovereign values residual cash flow less. Compensation for dilution therefore need not leave existing claims with the same value across states.

6.2 Advance Sales and Best-Price Protection

A seller that sells units or capacity claims before learning demand may later wish to cut prices and expand sales. In durable-good markets, those sales reduce earlier buyers' asset values, a loss the seller does not fully internalize. In advance-sale markets, buyers anticipate later price cuts and may wait, limiting the price the seller can charge initially. Restrictions on later sales can thus support initial sales, just as borrowing restrictions support initial debt, but they also limit the seller's response to demand.

[10] Switzerland's debt brake is one example (Federal Department of Finance, 2024). State-contingent debt has seen limited use (Igan, Kim, and Levy, 2022); examples link coupons to IMF or ratings tests in El Salvador (International Monetary Fund, 2025, p. 26, Box 2) and revenue and reporting targets in Sri Lanka (Democratic Socialist Republic of Sri Lanka, 2024), rather than new issuance.

Best-price protection allows later price cuts but requires the seller to refund the difference to earlier buyers. Because cutting prices now entails compensation, the seller has less incentive to expand output or cut prices. Full protection can restore dynamic consistency in durable-good markets, while shorter guarantees provide partial discipline (Butz, 1990). Under demand uncertainty, refunds let the seller adjust prices and sales while protecting earlier buyers (Png, 1991). Our model adds the screening requirement that the compensation must also induce truthful reporting of private information.

7 Conclusion

Financial flexibility allows firms to respond to financing needs that are difficult to anticipate or contract on. Under non-exclusive lending, however, future borrowing can dilute existing creditors. Initial debt therefore does more than finance investment: it creates the incentive problem that the borrowing restriction must address. Initial borrowing and future financial flexibility must be chosen jointly.

A simple debt limit balances creditor protection against access to refinancing. It permits excessive borrowing when liquidity needs are mild and restricts valuable borrowing when they are severe. An exclusive lender can instead price additional credit to account for dilution. Performance-sensitive debt and, when feasible, contingent prepayment reproduce this allocation by compensating existing creditors when the borrower raises new funds.

How much financial flexibility the borrower can retain therefore depends on the commitments lenders can make and enforce. When lenders cannot commit to future financing or administer state-contingent compensation, a debt limit offers simpler creditor protection. The resulting restriction on future borrowing is part of the cost of financing investment today.

References

Admati, A. R., P. M. Demarzo, M. F. Hellwig, and P. Pfleiderer (2018). The leverage ratchet effect. Journal of Finance 73(1), 145–198.

Aguiar, M., M. Amador, H. Hopenhayn, and I. Werning (2019). Take the short route: Equilibrium default and debt maturity. Econometrica 87(2), 423–462.

Alonso, V., C. Arroyo, O. Aydin, V. Balasundharam, H. R. Davoodi, G. Hegab, W. R. Lam, A. Nguyen, N. Salazar Ferro, G. Sher, A. Solovyeva, and N. Tchelishvili (2025). Fiscal rules dataset: 1985–2024. International Monetary Fund dataset.

Amador, M. and K. Bagwell (2013). The theory of optimal delegation with an application to tariff caps. Econometrica 81(4), 1541–1599.

Amador, M., I. Werning, and G. M. Angeletos (2006). Commitment vs. flexibility. Econometrica 74(2), 365–396.

Asquith, P., A. Beatty, and J. Weber (2005, December). Performance pricing in bank debt contracts. Journal of Accounting and Economics 40(1-3), 101–128.

Asriyan, V. and V. Vanasco (2024, March). Security design in non-exclusive markets with asymmetric information. Review of Economic Studies 91(2), 690–719.

Attar, A., C. Casamatta, A. Chassagnon, and J.-P. Décamps (2019, May). Multiple Lenders, Strategic Default, and Covenants. American Economic Journal: Microeconomics 11(2), 98–130.

Attar, A., T. Mariotti, and F. Salanié (2011). Nonexclusive competition in the market for lemons. Econometrica 79(6), 1869–1918.

Attar, A., T. Mariotti, and F. Salanié (2014). Nonexclusive competition under adverse selection. Theoretical Economics 9(1), 1–40.

Attar, A., T. Mariotti, and F. Salanié (2020). The social costs of side trading. Economic Journal 130(630), 1608–1622.

Avon Products, Inc. (2014). Form 10-K, Annual Report for the fiscal year ended December 31, 2013. https://www.sec.gov/Archives/edgar/data/8868/000000886814000013/avp10k2013.htm. U.S. Securities and Exchange Commission; accessed August 25, 2025.

Bagnoli, M. and T. Bergstrom (2005). Log-Concave Probability and Its Applications. Economic Theory 26(2), 445–469.

Baron, D. P. and D. Besanko (1984). Regulation and information in a continuing relationship. Information Economics and policy 1(3), 267–302.

Bensoussan, A., B. Chevalier-Roignant, and A. Rivera (2021, October). Does performance-sensitive debt mitigate debt overhang? Journal of Economic Dynamics and Control 131, 104203.

Bisin, A. and D. Guaitoli (2004). Moral Hazard and Nonexclusive Contracts. The RAND Journal of Economics 35(2), 306–328.

Bizer, D. S. and P. M. DeMarzo (1992). Sequential Banking. Journal of Political Economy 100(1), 41–61.

Black, F. and M. Scholes (1973). The pricing of options and corporate liabilities. Journal of Political Economy 81(3), 637–654.

Bradley, M. and M. R. Roberts (2015, June). The Structure and Pricing of Corporate Debt Covenants. Quarterly Journal of Finance 05(02), 1550001.

Bratton, W. W. (2016, October). Bond and loan covenants, theory and practice. Capital Markets Law Journal 11(4), 461–485.

Butz, D. A. (1990). Durable-good monopoly and best-price provisions. The American Economic Review 80(5), 1062–1076.

Castro-Pires, H. and H. Moreira (2021, July). Limited liability and non-responsiveness in agency models. Games and Economic Behavior 128, 73–103.

Chaigneau, P., A. Edmans, and D. Gottlieb (2021). When is (performance-sensitive) debt optimal?

DeMarzo, P., Z. He, and F. Tourre (2023, October). Sovereign debt ratchets and welfare destruction. Journal of Political Economy 131(10), 2825–2892.

DeMarzo, P. M. (2019). Presidential Address: Collateral and Commitment. Journal of Finance 74(4), 1587–1619.

DeMarzo, P. M. and Z. He (2021). Leverage dynamics without commitment. Journal of Finance 76(3), 1195–1250.

Democratic Socialist Republic of Sri Lanka (2024, November). Sri lanka announces the launch of international sovereign bonds exchange following cabinet of ministers approval. Ministry of Finance, Planning and Economic Development announcement.

Diamond, D. W., Y. Hu, and R. G. Rajan (2022). Liquidity, pledgeability, and the nature of lending. Journal of Financial Economics 143(3), 1275–1294.

Donaldson, J. R., D. Gromb, and G. Piacentino (2025). Conflicting priorities: A theory of covenants and collateral. Journal of Finance 80(3), 1739–1768.

Eckbo, B. E., K. Li, and W. Wang (2023). Loans to chapter 11 firms: contract design, repayment risk, and pricing. The Journal of Law and Economics 66(3), 465–509.

Fama, E. F. and M. Miller (1972). The Theory of Finance. Holt Rinehart & Winston.

Federal Department of Finance (2024). "Debt Brake". Swiss Confederation. Published April 10. https://www.efd.admin.ch/en/the-debt-brake.

Graham, J. R. (2022). Presidential address: Corporate finance and reality. The Journal of Finance 77(4), 1975–2049.

Halac, M. and P. Yared (2014). Fiscal rules and discretion under persistent shocks. Econometrica 82(5), 1557–1614.

Halac, M. and P. Yared (2018, August). Fiscal rules and discretion in a world economy. American Economic Review 108(8), 2305–2334.

Halac, M. and P. Yared (2022). Fiscal rules and discretion under limited enforcement. Econometrica 90(5), 2093–2127.

Halac, M. and P. Yared (2025). A theory of fiscal responsibility and irresponsibility. Journal of Political Economy 133(5), 1574–1620.

Hart, O. D. and J. Tirole (1988). Contract renegotiation and coasian dynamics. Review of Economic Studies 55(4), 509–540.

Hatchondo, J. C., L. Martinez, and C. Sosa-Padilla (2016). Debt dilution and sovereign default risk. Journal of Political Economy 124(5), 1383–1422.

Holmström, B. (1984). On the theory of delegation. In M. Boyer and R. E. Kihlstrom (Eds.), Bayesian Models in Economic Theory, pp. 115–141. Amsterdam: North-Holland.

Hu, Y. and F. Varas (2025). Intermediary financing without commitment. Journal of Financial Economics 167, 104025.

Hu, Y., F. Varas, and C. Ying (2026). Debt maturity management. The Review of Financial Studies.

Igan, D., T. Kim, and A. Levy (2022, January). The premia on state-contingent sovereign debt instruments. BIS Working Papers 988, Bank for International Settlements.

International Monetary Fund (2025). El Salvador: Request for Extended Arrangement under the Extended Fund Facility—Press Release; Staff Report; Staff Statement; and Statement by the Executive Director for El Salvador. IMF Country Report 25/58, International Monetary Fund, Washington, DC. March. https://www.imf.org/-/media/files/publications/cr/2025/english/1slvea2025001-print-pdf.pdf.

Jensen, M. C. (1986). Agency costs of free cash flow, corporate finance, and takeovers. The American economic review 76(2), 323–329.

Kahn, C. M. and D. Mookherjee (1998). Competition and Incentives with Nonexclusive Contracts. The RAND Journal of Economics 29(3), 443–465.

Manso, G., B. Strulovici, and A. Tchistyi (2010). Performance-sensitive debt. The Review of Financial Studies 23(5), 1819–1854.

Peters, M. and B. Szentes (2012). Definable and contractible contracts. Econometrica 80(1), 363–411.

Png, I. P. L. (1991). Most-favored-customer protection versus price discrimination over time. Journal of Political Economy 99(5), 1010–1028.

Segal, I. (1999). Contracting with Externalities. Quarterly Journal of Economics 114(2), 337–388.

Simpson Thacher & Bartlett LLP (2022). Leveraged finance 101: A covenant handbook. Technical report, Simpson Thacher & Bartlett LLP. Available at https://www.stblaw.com/about-us/publications/view/2022/09/12/simpson-thacher-publishes-em-leveraged-finance-101-a-covenant-handbook-em.

Smith, C. W. and J. B. Warner (1979, June). On financial contracting. Journal of Financial Economics 7(2), 117–161.

Szentes, B. (2015). Contractible contracts in common agency problems. The Review of Economic Studies 82(1 (290)), 391–422.

Appendix

A Non-Exclusive Lending with Simple Debt Contracts

Auxiliary Results

We begin with preliminary results used in the derivations that follow. First, we state the first-order condition for the cutoff in the debt-limit problem (15). Let $\lambda \geq 0$ be the multiplier on the financing constraint and set $\delta = 1/(1+\lambda)$. Dividing the Lagrangian by $1+\lambda$ and differentiating with respect to θ_L shows that, for $\theta_L > \underline{\theta}$, the derivative has the opposite sign to the left-hand side of

$$x_0 \frac{q'(x_1^f(\theta_L))}{q(x_1^f(\theta_L))} + \delta \mathbb{E}[\theta_L - \theta \mid \theta \leq \theta_L] = 0. \tag{28}$$

We refer to (28) as the cutoff condition. It holds with equality at every interior cutoff, and at $\delta = 1$ it reduces to (16) in Proposition 1. The derivation can be found in the Supplemental Appendix OA.1, where it is used to prove that proposition.

Several proofs use some properties about the inverse-demand elasticity $\varepsilon(x) = -xq'(x)/q(x)$, the efficient debt level $x_1^*(\theta)$, the conditional mean gap, and the response of flexible debt to initial debt and to the type, which we collect in the following lemmas. Let $\Gamma(x) = [x\varepsilon'(x) - \varepsilon(x)]/\varepsilon(x)^2$, let $x^m \coloneqq \arg\max_{x \in [0,Y]} xq(x)$, and, for $\theta > \underline{\theta}$, let

$$\rho(\theta) \coloneqq \mathbb{E}[\theta - \tilde{\theta} \mid \tilde{\theta} \leq \theta].$$

Lemma 3 (Elasticity, efficient debt, and the debt-limit bound)**.** *Under Assumptions 1 and 3:*

1. *$\varepsilon(x)$ and $\varepsilon(x)/x = -q'(x)/q(x)$ are strictly increasing on $(0, Y)$, and $\Gamma(x) \geq 0$.*
2. *$x_1^*(\theta)$ is the unique solution of $\varepsilon(x) = 1 - \theta$. Hence x_1^* is continuously differentiable and strictly decreasing, $x^m = x_1^*(0)$ is the unique solution of $\varepsilon(x) = 1$, and $xq(x)$ is strictly increasing on $[0, x^m]$.*
3. *$\rho(\theta) = \int_{\underline{\theta}}^{\theta} G(s)\,ds/G(\theta)$, $\rho'(\theta) = 1 - g(\theta)\rho(\theta)/G(\theta) \geq 0$, and $\rho(\theta) < \theta - \underline{\theta}$.*

4. *Let* $x_L = x_1^f(\theta_L, x_0)$ *with* $\theta_L \in (\underline{\theta}, 1)$ *satisfy the cutoff condition* (28) *for some* $\delta \in (0, 1]$. *Then*

$$\varepsilon(x_L) = 1 - \theta_L + \delta\rho(\theta_L) < 1 - \underline{\theta}, \qquad \textit{so} \qquad x_L < x_1^*(\underline{\theta}) < x^m.$$

Proof. For part (1), $q' < 0$ and $q'' \leq 0$ (Lemma 1 and Assumption 3) give $\frac{d}{dx}\big(-q'(x)/q(x)\big) = \big(q'(x)^2 - q''(x)q(x)\big)/q(x)^2 > 0$, so $\varepsilon(x)/x$ is positive and strictly increasing, and so is $\varepsilon(x)$. The same computation gives $x\varepsilon'(x) - \varepsilon(x) = x^2\big(q'(x)^2 - q''(x)q(x)\big)/q(x)^2 > 0$, so $\Gamma \geq 0$.

For part (2), $V_2' = -q$ gives $\partial W_1(x,\theta)/\partial x_1 = q(x)\big[1 - \theta - \varepsilon(x)\big]$. Since ε is strictly increasing with $\varepsilon(0) = 0$ and $\varepsilon(x) \to \infty$ as $x \to Y$, this derivative single-crosses zero from above, at the unique root $x_1^*(\theta)$ of $\varepsilon(x) = 1 - \theta$, and the implicit-function theorem gives $x_1^{*\prime}(\theta) = -1/\varepsilon'(x_1^*(\theta)) < 0$. At $\theta = 0$, $W_1(x, 0) = xq(x)$, so $x^m = x_1^*(0)$ is the unique root of $\varepsilon(x) = 1$ and $xq(x)$ increases on $[0, x^m]$.

For part (3), integration by parts gives $\int_{\underline{\theta}}^{\theta} G(s)\,ds = G(\theta)\rho(\theta)$, and differentiating this identity gives the formula for ρ'. Log-concavity of G implies log-concavity of $\int_{\underline{\theta}}^{\theta} G(s)\,ds$ (Bagnoli and Bergstrom, 2005), which is $\rho'(\theta) \geq 0$. Since $g > 0$, $\mathbb{E}[\tilde{\theta} \mid \tilde{\theta} \leq \theta] > \underline{\theta}$, so $\rho(\theta) < \theta - \underline{\theta}$.

For part (4), divide the flexible-debt condition (14) by $q(x_L)$ to get $(1 - x_0/x_L)\varepsilon(x_L) = 1 - \theta_L$, and write the cutoff condition (28) as $(x_0/x_L)\varepsilon(x_L) = \delta\rho(\theta_L)$. Adding the two relations gives the equality. The inequality follows from $\delta \leq 1$ and part (3), and parts (1) and (2) then give $x_L < x_1^*(\underline{\theta}) < x_1^*(0) = x^m$. □

Lemma 4 (Flexible-debt derivatives)**.** *For feasible* x_0 *and* $\theta \in \Theta$, *the flexible-debt condition* (14) *is equivalent to*

$$\Big(1 - \frac{x_0}{x_1^f(\theta, x_0)}\Big)\varepsilon\big(x_1^f(\theta, x_0)\big) = 1 - \theta. \tag{29}$$

Under Assumption 3, x_1^f *is continuously differentiable, with*

$$\frac{\partial x_1^f(\theta, x_0)}{\partial x_0} = \frac{1}{1 + (1-\theta)\Gamma\big(x_1^f(\theta, x_0)\big)} \in (0, 1],$$
$$\frac{\partial x_1^f(\theta, x_0)}{\partial \theta} = -\frac{x_1^f(\theta, x_0)}{\varepsilon\big(x_1^f(\theta, x_0)\big)}\,\frac{\partial x_1^f(\theta, x_0)}{\partial x_0} < 0.$$

Proof. Dividing (14) by $q(x_1^f) > 0$ gives (29). Apply the implicit-function theorem to $F(x, x_0, \theta) = (1 - x_0/x)\varepsilon(x) - (1 - \theta)$, whose root in x is $x_1^f(\theta, x_0)$. Then $F_{x_0} = -\varepsilon(x)/x$, $F_\theta = 1$, and, after substituting $x_0/x = 1 - (1-\theta)/\varepsilon(x)$ from (29),

$$F_x = \frac{x_0}{x^2}\varepsilon(x) + \Big(1 - \frac{x_0}{x}\Big)\varepsilon'(x) = \frac{\varepsilon(x)}{x}\big[1 + (1-\theta)\Gamma(x)\big] > 0,$$

where the sign uses $\Gamma \geq 0$ from Lemma 3. The same bound gives $\partial x_1^f/\partial x_0 \leq 1$. □

Proof of Proposition 3

We first derive how the borrower's value changes with initial debt, then use this derivative to show that financing binds and to characterize the optimal contract.

Since $x_0 q(x_0) \geq I$ if and only if $x_0 \in [x_0^{\min}, x_0^{\max}]$ (Section 3.3), problem (19) is

$$\max_{x_0 \in [x_0^{\min}, x_0^{\max}]} W_0^L(x_0). \tag{30}$$

Write $\theta \vee \theta_L := \max\{\theta, \theta_L\}$. First consider $I = I^{\max}$, when the two roots coincide at $x_1^*(0)$. Any cutoff $\theta_L < 1$ induces additional borrowing with positive probability and lowers the expected price of initial debt, so $x_0 \mathbb{E}[q(x_1^f(\theta \vee \theta_L, x_0))] < x_0 q(x_0) = I^{\max}$. Only $\theta_L = 1$ is therefore feasible, which gives the first case of the proposition. For the remainder of the proof, suppose $I < I^{\max}$, so $x_0^{\min} < x_0^{\max}$.

For fixed initial debt, Lemma 2 gives implemented total debt as $x_1^f(\theta \vee \theta_L, x_0)$. The borrower's value is therefore

$$\begin{aligned} W_0^L(x_0) = \max_{\theta_L \in \Theta} \mathbb{E}\left[W_1\left(x_1^f(\theta \vee \theta_L, x_0), \theta\right)\right] \\ \text{subject to} \quad x_0 \mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L, x_0)\right)\right] \geq I. \end{aligned} \tag{31}$$

Let $W_1'(x_1, \theta)$ denote the derivative with respect to x_1, and write $x_1^L = x_1^f(\theta_L, x_0)$ for the debt limit. With $\lambda = \delta^{-1} - 1$ denoting the multiplier on the financing constraint,

the envelope theorem gives

$$W_0^{L\prime}(x_0) = \mathbb{E}\left[W_1'\left(x_1^f(\theta \vee \theta_L, x_0), \theta\right) \frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0}\right] + \lambda \mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L, x_0)\right) + x_0 q'\left(x_1^f(\theta \vee \theta_L, x_0)\right) \frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0}\right], \quad (32)$$

where $\partial x_1^f(\theta, x_0)/\partial x_0 \in (0, 1]$ by Lemma 4. By the flexible-debt condition (14), $W_1'(x_1^f(\theta \vee \theta_L, x_0), \theta) = x_0 q'(x_1^f(\theta \vee \theta_L, x_0)) + (\theta_L - \theta) q(x_1^L) \mathbf{1}_{\{\theta \leq \theta_L\}}$. Substituting this identity and, for $\theta_L < 1$, the cutoff condition (28) into (32) cancels the terms involving $\partial x_1^f(\theta_L, x_0)/\partial x_0$ and yields

$$W_0^{L\prime}(x_0) = \delta^{-1} \mathbb{E}\left[(1-\delta) q(x_1^f(\theta \vee \theta_L, x_0)) + x_0 q'(x_1^f(\theta, x_0)) \frac{\partial x_1^f(\theta, x_0)}{\partial x_0} \mathbf{1}_{\{\theta > \theta_L\}}\right]. \quad (33)$$

If instead $\theta_L = 1$, total debt equals x_0 and $W_0^L(x_0) = W_1(x_0, \mathbb{E}[\theta])$.

We now show that financing binds at every solution x_0^* of (30), with associated cutoff θ_L^*. Suppose, for contradiction, that financing is slack. Then x_0^* lies strictly between the two roots, and complementary slackness gives $\delta = 1$. If $\theta_L^* < 1$, equation (33) gives

$$W_0^{L\prime}(x_0^*) = x_0^* \mathbb{E}\left[q'\left(x_1^f(\theta, x_0^*)\right) \frac{\partial x_1^f(\theta, x_0^*)}{\partial x_0} \mathbf{1}_{\{\theta > \theta_L^*\}}\right] < 0,$$

because $q' < 0$, $\partial x_1^f/\partial x_0 > 0$, and $\mathbb{P}(\theta > \theta_L^*) > 0$. A small reduction in initial debt is then feasible and raises the borrower's value. If $\theta_L^* = 1$, Proposition 1 gives $x_0^* \geq x_1^*(\mathbb{E}[\theta])$. When the inequality is strict, a small reduction in initial debt preserves feasibility and raises $W_1(x_0, \mathbb{E}[\theta])$. When $x_0^* = x_1^*(\mathbb{E}[\theta])$, financing remains slack for initial debt slightly below x_0^*, where Proposition 1 gives $\theta_L < 1$ and the display above gives $W_0^{L\prime} < 0$. Each case contradicts optimality, so financing binds.

Binding financing determines the two cases of the proposition. If $x_0^{\min} \geq x_1^*(\mathbb{E}[\theta])$, Proposition 1 gives $\theta_L = 1$ throughout the feasible interval, so $W_0^L(x_0) = W_1(x_0, \mathbb{E}[\theta])$ decreases on that interval and $x_0^* = x_0^{\min}$. If $x_0^{\min} < x_1^*(\mathbb{E}[\theta])$, feasibility at either root requires $\theta_L = 1$, and a small move from either root toward $x_1^*(\mathbb{E}[\theta])$ is feasible and raises $W_1(x_0, \mathbb{E}[\theta])$; for the upper root, this uses $x_0^{\max} > x_1^*(0) > x_1^*(\mathbb{E}[\theta])$. Optimal initial debt therefore lies strictly between the two roots. Binding financing then rules

out $\theta_L^* = 1$, which would require $x_0^* q(x_0^*) = I$ and place initial debt at a root. Hence, $\theta_L^* < 1$, and setting (33) equal to zero gives the initial-debt condition in (20). Together with binding financing and the cutoff condition (28), this proves the second case of the proposition.

Proof of Proposition 4

We first characterize the portion of the binding financing schedule on which an optimum must lie. We then show that the borrower's value is single-peaked along this portion of the schedule. Proposition 3 establishes the result when $x_1^*(\mathbb{E}[\theta]) \leq x_0^{\min}$. Suppose instead that $x_1^*(\mathbb{E}[\theta]) > x_0^{\min}$. The same proposition implies that financing binds, $\theta_L < 1$, and initial debt lies strictly between the two financing roots. In particular, $I < I^{\max}$.

Write $x_L = x_1^f(\theta_L, x_0)$ for the debt limit. Define the expected debt price and financing surplus by

$$Q(x_0, \theta_L) \coloneqq \mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L, x_0)\right)\right], \qquad \mathcal{F}(x_0, \theta_L) \coloneqq x_0 Q(x_0, \theta_L) - I.$$

Financing capacity need not be monotone in initial debt, so $\mathcal{F}(x_0, \theta_L) = 0$ can have several roots. We show that every optimum lies in the region $x_L < x^m$, where there is at most one root for each cutoff.

For $\theta_L \in (\underline{\theta}, 1)$,

$$\mathcal{F}_{\theta_L}(x_0, \theta_L) = x_0 G(\theta_L) q'(x_L) \frac{\partial x_1^f(\theta_L, x_0)}{\partial \theta_L} > 0.$$

The cutoff $\theta_L = \underline{\theta}$ cannot be optimal: raising it slightly relaxes financing, and it raises the borrower's payoff because the payoff derivative has the sign opposite to the left-hand side of the cutoff condition (28) at $\delta = 1$, which is negative near $\underline{\theta}$ since $\rho(\theta_L) \to 0$ and $q' < 0$.

Every optimum therefore has $\theta_L \in (\underline{\theta}, 1)$. Since $\mathcal{F}_{\theta_L} > 0$, the joint first-order conditions apply even if $\mathcal{F}_{x_0} = 0$. The cutoff condition holds with $\delta \leq 1$, so Lemma 3 gives $x_L < x^m$, including when the financing constraint binds with $\delta = 1$.

By (29), $x_L = x^m$ when $x_0 = \theta_L x^m$. Since flexible debt strictly increases with

initial debt (Lemma 4), $x_L < x^m$ is equivalent to $x_0 < \theta_L x^m$. Throughout this region,

$$x_0 \leq x_1^f(\theta \vee \theta_L, x_0) \leq x_L < x^m.$$

Since every implemented debt level has elasticity below one and $\partial x_1^f/\partial x_0 \leq 1$ by Lemma 4, differentiating financing capacity gives

$$\mathcal{F}_{x_0}(x_0, \theta_L) = \mathbb{E}\left[q\left(x_1^f(\theta \vee \theta_L, x_0)\right) + x_0 q'\left(x_1^f(\theta \vee \theta_L, x_0)\right) \frac{\partial x_1^f(\theta \vee \theta_L, x_0)}{\partial x_0}\right] > 0.$$

Indeed, the absolute value of the second term relative to the first is the product of x_0/x_1^f, the elasticity, and $\partial x_1^f/\partial x_0$, which is strictly below one. Thus, for each cutoff, binding financing has at most one root in this region.

At $\theta_L = 1$, that root is $x_0^{\min}$. The implicit-function theorem continues it as a schedule $x_0 = \chi(\theta_L)$ unless x_L reaches x^m. At that boundary, $x_0 = \theta_L x^m$, so the cutoff condition would require $\delta = \theta_L/\rho(\theta_L) > 1$. The schedule therefore exists throughout the relevant interval where the cutoff condition (28) gives $\delta \leq 1$, and

$$\chi(\theta_L) Q(\chi(\theta_L), \theta_L) = I, \qquad \chi'(\theta_L) = -\frac{\mathcal{F}_{\theta_L}}{\mathcal{F}_{x_0}} < 0,$$

where the derivatives are evaluated at $(\chi(\theta_L), \theta_L)$. At the smallest cutoff in this interval, $\delta = 1$.

Lemma 5 (Debt-price sensitivity to initial debt)**.** *If* Γ *is nonincreasing, then, for each type* θ*,*

$$-q'(x_1^f(\theta, x_0)) \frac{\partial x_1^f(\theta, x_0)}{\partial x_0}$$

is nondecreasing in x_0*.*

Proof. Writing $x = x_1^f(\theta, x_0)$, Lemma 4 gives

$$-q'(x)\frac{\partial x}{\partial x_0} = \frac{-q'(x)}{1 + (1-\theta)\Gamma(x)}.$$

Assumption 3 implies that $-q'(x)$ is nondecreasing, while the positive denominator is nonincreasing. The ratio is therefore nondecreasing in x, which increases with x_0 by Lemma 2. □

Along the binding schedule, write $x_L = x_1^f(\theta_L, \chi(\theta_L))$ and define

$$N(x_0, \theta_L) \coloneqq -\mathbb{E}\left[q'(x_1^f(\theta, x_0))\frac{\partial x_1^f(\theta, x_0)}{\partial x_0}\mathbf{1}_{\{\theta > \theta_L\}}\right].$$

Using $Q(\chi(\theta_L), \theta_L) = I/\chi(\theta_L)$, the two first-order conditions agree if and only if $\mathcal{S}(\theta_L) = 0$, where

$$\mathcal{S}(\theta_L) \coloneqq 1 - \frac{\chi(\theta_L)^2 N(\chi(\theta_L), \theta_L)}{I} - \frac{\chi(\theta_L)}{x_L}\frac{\varepsilon(x_L)}{\rho(\theta_L)}.$$

The first two terms give δ from the initial-debt condition (33); the last term gives δ from the debt-limit condition (28).

Lemma 6 (Single crossing of $\mathcal{S}$)**.** *If Γ is nonincreasing, then $\mathcal{S}(\theta_L)$ is strictly increasing on the relevant cutoff interval.*

Proof. Since $\chi'(\theta_L) < 0$, the debt limit $x_L = x_1^f(\theta_L, \chi(\theta_L))$ decreases with θ_L. Write the last term of $\mathcal{S}$ as $\chi(\theta_L)\left[\varepsilon(x_L)/x_L\right]/\rho(\theta_L)$. By Lemma 3, $\varepsilon(x)/x$ increases with x, so the middle factor decreases as θ_L rises, and $\rho(\theta_L)$ is nondecreasing. Together with $\chi'(\theta_L) < 0$, these signs imply that the last term decreases.

For the second term, Lemma 5 implies that the integrand defining N is nonincreasing as $\chi(\theta_L)$ falls, and the integration region $\{\theta > \theta_L\}$ shrinks, so $N(\chi(\theta_L), \theta_L)$ decreases. Since $N > 0$ for $\theta_L < 1$ and χ strictly decreases, $\chi^2 N$ strictly decreases. Hence, $\mathcal{S}$ strictly increases. □

Conclusion. The envelope calculation in the proof of Proposition 3 gives the derivative of the borrower's value along the binding schedule as $\chi'(\theta_L)Q(\chi(\theta_L), \theta_L)\mathcal{S}(\theta_L)/\delta$. Since $Q, \delta > 0$ and $\chi' < 0$, value increases when $\mathcal{S} < 0$ and decreases when $\mathcal{S} > 0$.

At the smallest relevant cutoff, the debt-limit condition gives $\delta = 1$, so the last term of $\mathcal{S}$ equals one. Since $N > 0$, we have $\mathcal{S} < 0$ there. At $\theta_L = 1$, $N = 0$ and $\chi(1) = x_0^{\min}$, so

$$\mathcal{S}(1) = 1 - \frac{\varepsilon(x_0^{\min})}{1 - \mathbb{E}[\theta]} > 0.$$

The strict inequality follows from $x_0^{\min} < x_1^*(\mathbb{E}[\theta])$, strict monotonicity of ε, and $\varepsilon(x_1^*(\mathbb{E}[\theta])) = 1 - \mathbb{E}[\theta]$.

By Lemma 6, $\mathcal{S}$ crosses zero exactly once. The borrower's value rises up to this crossing and falls thereafter. Every optimum lies on the binding schedule considered

above, so the unique solution to the necessary conditions is the unique maximizer of (19). □

Proof of Proposition 5

Proof. We differentiate the financing and optimality conditions to show that initial debt and the cutoff increase with I, then use the same system to show that the debt limit decreases. We use the notation from the proof of Proposition 4, with $x_L = x_1^f(\theta_L, x_0)$. Before imposing the financing constraint, write the difference between the discount factors implied by (20) and (28) as

$$\mathcal{S}(x_0, \theta_L) = 1 - \frac{x_0 N(x_0, \theta_L)}{Q(x_0, \theta_L)} - \frac{x_0}{x_L}\frac{\varepsilon(x_L)}{\rho(\theta_L)}. \tag{34}$$

On the binding schedule, $\mathcal{S}(\chi(\theta_L), \theta_L)$ equals $\mathcal{S}(\theta_L)$ in the preceding proof. Partial derivatives below refer to the two-variable form. At every interior zero,

$$\delta = 1 - \frac{x_0 N}{Q} = \frac{x_0 \varepsilon(x_L)}{x_L \rho} \in (0, 1). \tag{35}$$

Lemma 3 then gives

$$x_L < x_1^*(\underline{\theta}) < x^m, \tag{36}$$

so every interior zero lies in the low-debt region.

Initial debt and the cutoff. Since differentiating N directly would require q''', change variables from type to debt, which makes N continuously differentiable given $q \in C^2$ and $g \in C^1$:

$$N(x_0, \theta_L) = \int_{x_0}^{x_L} \frac{q'(z)^2}{q(z)} g\left(1 + (z - x_0)\frac{q'(z)}{q(z)}\right) dz.$$

Lemma 5 gives $N_{x_0} \geq 0$, while reducing the set of unconstrained borrowers gives

$$N_{\theta_L} = g(\theta_L) q'(x_L) \frac{\partial x_1^f(\theta_L, x_0)}{\partial x_0} < 0.$$

The expected price Q decreases with x_0 and increases with θ_L. Moreover, $\varepsilon(x)/x$ strictly increases with x and ρ is nondecreasing (Lemma 3). Together with Lemma

2, these signs give $\mathcal{S}_{x_0} < 0 < \mathcal{S}_{\theta_L}$ at every interior optimum.

The financing surplus $\mathcal{F}(x_0, \theta_L) = x_0 Q(x_0, \theta_L) - I$ satisfies $\mathcal{F}_{x_0}, \mathcal{F}_{\theta_L} > 0$ in this region, as shown in the preceding proof, so the Jacobian of the system $\mathcal{F} = \mathcal{S} = 0$ is nonsingular. Since $\mathcal{F}_I = -1$ and $\mathcal{S}_I = 0$, implicit differentiation gives $\mathcal{S}_{x_0}\, dx_0/dI + \mathcal{S}_{\theta_L}\, d\theta_L/dI = 0$, so the two derivatives share a sign, and $\mathcal{F}_{x_0}\, dx_0/dI + \mathcal{F}_{\theta_L}\, d\theta_L/dI = 1$ then makes both strictly positive.

The debt limit. Higher initial debt and a higher cutoff have opposing effects on x_L, so we first compare contracts holding the debt limit fixed. Solving (29) for the cutoff gives $\theta_L = 1 - (x_L - x_0)\varepsilon(x_L)/x_L$. At a fixed debt limit, the cutoff therefore rises with initial debt at rate $\varepsilon(x_L)/x_L$. Constrained borrowers face the same debt price, so the expected price changes only through the higher total debt chosen by unconstrained borrowers:

$$Q_{x_0} + \frac{\varepsilon(x_L)}{x_L} Q_{\theta_L} = -N.$$

At $\mathcal{S} = 0$, differentiate (34) with respect to x_0 while holding x_L fixed. Using this identity, (35), and $\rho' = 1 - g(\theta_L)\rho/G(\theta_L)$ gives

$$\begin{aligned} &x_0 \left(\mathcal{S}_{x_0} + \frac{\varepsilon(x_L)}{x_L} \mathcal{S}_{\theta_L} \right) \\ &\quad = -2(1-\delta) - \frac{x_0^2 N_{x_0}}{Q} - \delta^2 \frac{g(\theta_L)\rho}{G(\theta_L)} \left(1 - \frac{\rho G(\theta_L) q(x_L)}{Q} \frac{\partial x_1^f(\theta_L, x_0)}{\partial x_0} \right) < 0. \end{aligned} \tag{37}$$

The sign follows from $N_{x_0} \geq 0$, $\delta, \rho \in (0,1)$, $G(\theta_L)q(x_L) \leq Q$, and $0 < \partial x_1^f/\partial x_0 \leq 1$.

Solving the two-equation system above and applying the derivative ratio in Lemma 4 to $x_L = x_1^f(\theta_L, x_0)$ yields

$$\frac{dx_L^{n*}}{dI} = -\frac{\partial x_1^f/\partial \theta_L}{\mathcal{F}_{x_0}\mathcal{S}_{\theta_L} - \mathcal{F}_{\theta_L}\mathcal{S}_{x_0}} \left(\mathcal{S}_{x_0} + \frac{\varepsilon(x_L)}{x_L} \mathcal{S}_{\theta_L} \right) < 0.$$

All derivatives are evaluated at the optimum. The denominator is positive, $\partial x_1^f/\partial \theta_L < 0$, and the parenthesis is negative by (37). By uniqueness, these local solutions describe the optimum throughout the investment range in the proposition. □